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Specificity of peripheral nerve regeneration: Interactions at the axon level

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ABSTRACT

Peripheral nerves injuries result in paralysis, anesthesia and lack of autonomic control of the affected body areas. After injury, axons distal to the lesion are disconnected from the neuronal body and degenerate, leading to denervation of the peripheral organs. Wallerian degeneration creates a microenvironment distal to the injury site that supports axonal regrowth, while the neuron body changes in phenotype to promote axonal regeneration. The significance of axonal regeneration is to replace the degenerated distal nerve segment, and achieve reinnervation of target organs and restitution of their functions. However, axonal regeneration does not always allows for adequate functional recovery, so that after a peripheral nerve injury, patients do not recover normal motor control and fine sensibility. The lack of specificity of nerve regeneration, in terms of motor and sensory axons regrowth, pathfinding and target reinnervation, is one the main shortcomings for recovery. Key factors for successful axonal regeneration include the intrinsic changes that neurons suffer to switch their transmitter state to a pro-regenerative state and the environment that the axons find distal to the lesion site. The molecular mechanisms implicated in axonal regeneration and pathfinding after injury are complex, and take into account the cross-talk between axons and glial cells, neurotrophic factors, extracellular matrix molecules and their receptors. The aim of this review is to look at those interactions, trying to understand if some of these molecular factors are specific for motor and sensory neuron growth, and provide the basic knowledge for potential strategies to enhance and guide axonal regeneration and reinnervation of adequate target organs.

Contents

Abbreviations: AKT/PKB, protein kinase B; BDNF, brain derived neurotrophic factor; cAMP, cyclic adenosine monophosphate; CAMs, cell adhesion molecules; Cdc42, cell division control protein 42 homolog; cGMP, cyclic guanosine monophosphate; ChAT, choline acetyltransferase; CGRP, calcitonin gene-related peptide; CNTF, ciliary neurotrophic factor; CSPG, chondroitin sulfate proteoglycan; DRG, dorsal root ganglia; ECM, extracellular matrix; ERK, extracellular signal-regulated kinase; FGF, fibroblast growth factor; FGFR, fibroblast growth factor receptor; GAP, growth associated protein; GDNF, glial cell-derived neurotrophic factor; GFR, glial cell derived neurotrophic factor family receptor; GGF, glial growth factor; GSK3β, glycogen synthase kinase 3 beta; HGF/SF, hepatocyte growth factor/scatter factor; HNK1, human natural killer 1; HSPG, heparan sulfate protoglycan; IB4, isolectin B4; IGF-1, insulin-like growth factor 1; IL-6, interleukin 6; JNK, c-Jun N-terminal kinase; LIF, leukemia inhibitory factor; mTOR, mammalian target of rapamycin; MAP, microtubule-associated protein; MAPK, Mitogen-activated protein (MAP) kinase; NCAM, neuronal cell adhesion molecule; NGF, nerve growth factor; NRG1, neuregulin 1; NT-3, neurotrophin 3; OPN, osteopontin; P38, mitogen-activated protein kinase 38; P75, low-affinity nerve growth factor receptor/P75 neurotrophin receptor; PKA, protein kinase A; PMR, preferential motor reinnervation; PNS, peripheral nervous system; PTEN, phosphatase and tensin homolog; PTN, pleiotrophin; RET, proto-oncogene-receptor tyrosine kinase; Rho, ras homolog gene family; Sema 3A, semaphoring 3A; SP, substance P; STAT3, signal transducer and activator of transcription 3; TGFα-1, transforming growth factor alpha 1; Thy-1/CD90, cluster of differentiation 90; TrK, receptor tyrosine kinase; TSC, tuberous sclerosis protein; VEGF, vascular endothelial growth factor.

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1. Introduction

Injuries to the peripheral nerves result in loss of motor, sensory and autonomic functions conveyed by the involved nerves. After a nerve injury, transected fibers distal to the lesion are disconnected from the neuronal body and undergo Wallerian degeneration, thus, leaving the peripheral organs denervated. In parallel, a series of molecular and cellular changes known as retrograde reaction and chromatolysis occur at the soma of axotomized neurons. Wallerian degeneration serves to create a microenvironment distal to the injury site that favors axonal regrowth, while retrograde reaction leads to metabolic changes necessary for regeneration and axonal elongation. The functional significance of axonal regeneration is to replace the distal nerve segment lost during degeneration, allowing reinnervation of target organs and restitution of their corresponding functions. Through this sequence of events, injured axons of the peripheral nervous system are able to regenerate and reinnervate their target organs.

After axonotmesis, where the connective sheaths of the nerve are preserved and only the axons are injured, functional recovery is usually good. In contrast, after neurotmesis (nerve transection), when the endoneurial tubes loss their continuity, axons are often misdirected and reinnervate incorrect target organs even if refined repair is applied (Bodine-Fowler et al., 1997; Molander and Aldskogius, 1992; Valero-Cabre and Navarro, 2002). Thus, although the amount of axonal regeneration can be considerably high, the lack of selectivity of axon-target reconnection leads to a poor functional recovery. Indeed, only a low percentage of adult patients regain normal function after complete transection and surgical repair of a major peripheral nerve. Appropriate and inappropriate targets can be reinnervated by axotomized neurons. For example, efferent motor axons may be misdirected to sensory end organs, and cutaneous afferents to motor endplates or sensory end organs of inappropriate modality or location. Thus, function will be degraded or permanently lost, depending on the severity of mismatch. As surgical nerve repair techniques cannot be further refined, there is a need for new and improved strategies to enhance specific axon regeneration following nerve injuries (Lundborg, 2003). Tissue specificity, i.e. preferential growth of axons towards a distal nerve stump rather than to other tissues, has been documented (Politis et al., 1982). Fascicular specificity, the preferential regeneration through the original nerve fascicle, was suggested in early studies but not confirmed by further investigations. Target organ specificity, or adequate reinnervation of each type of end organ (muscle, sensory receptor, . . .) by axons that originally served that organ, is less than perfect, although preferential reinnervation has been observed (Fig. 1). However, the mechanisms through which motor and sensory axons specifically reinnervate their corresponding targets are still poorly understood. Some authors defend a preferential muscle reinnervation by motor axons, the so-called preferential motor reinnervation (Brushart, 1988, 1993; Brushart et al., 1998). Pruning of the axons that reinnervated an erroneous target may contribute to improve the specificity of regeneration (Madison et al., 1996). Expression of the peptide L2/HNK1 (Martini et al., 1992) in motor but not sensory Schwann cells, or the presence of NCAM and polysialic acid in the regenerative motor axons (Franz et al., 2005) can also mediate this preferential motor reinnervation, although other authors argue that the key point for preferential attraction of axons to their targets is the expression of trophic factors by the own target organ and the distal stump (Madison et al., 2007).

Since the mechanisms that control axon regeneration are diverse and complex, it is important to take all of them into account before designing new strategies that may improve specific reinnervation. After the lesion, the injured neurons suffer important changes to switch their neuro-transmitter state to a pro-regenerative state (Plunet et al., 2002). The environment that the axons find distal to the lesion site is also a determinant for successful regeneration. Neurotrophic factors secreted by Schwann cells or target organs modulate axonal attraction, and the creation of a favorable pathway in the distal nerve stump allows the growing axons to adhere and elongate. The cross-talk between axons and their immediate environment is thus essential in determining the intrinsic capacities for regeneration and reinnervation. Therefore, it is important to deeper understand the molecular interactions between neurotrophic factors, their receptors, adhesion molecules and extracellular matrix elements in the regenerative process. These clues are involved in cell chemotaxis, migration and axonal elongation, and their signaling pathways can promote cell survival or apoptosis. The focus of this work is to review those interactions, trying to understand whether any factors can be specific for motor and sensory axons, and how can they help to induce specific reinnervation of target organs.

2. Neuronal changes triggered by axonal injury

Neurons have an intrinsic growth capacity during the embryonic stage, which is repressed upon the adult transition to allow proper synaptic development. However, after axotomy, neurons switch again from a transmission state to a growth state, with changes in the expression of genes that encode for transcription factors (Herdegen et al., 1991; Leah et al., 1991; Schwaiger et al., 2000), which in turn regulate the expression of genes involved in cell survival and neurite outgrowth (for reviews see Navarro et al., 2007; Raivich and Makwana, 2007). This switch is essential in the capacity of neurons to regenerate; therefore the

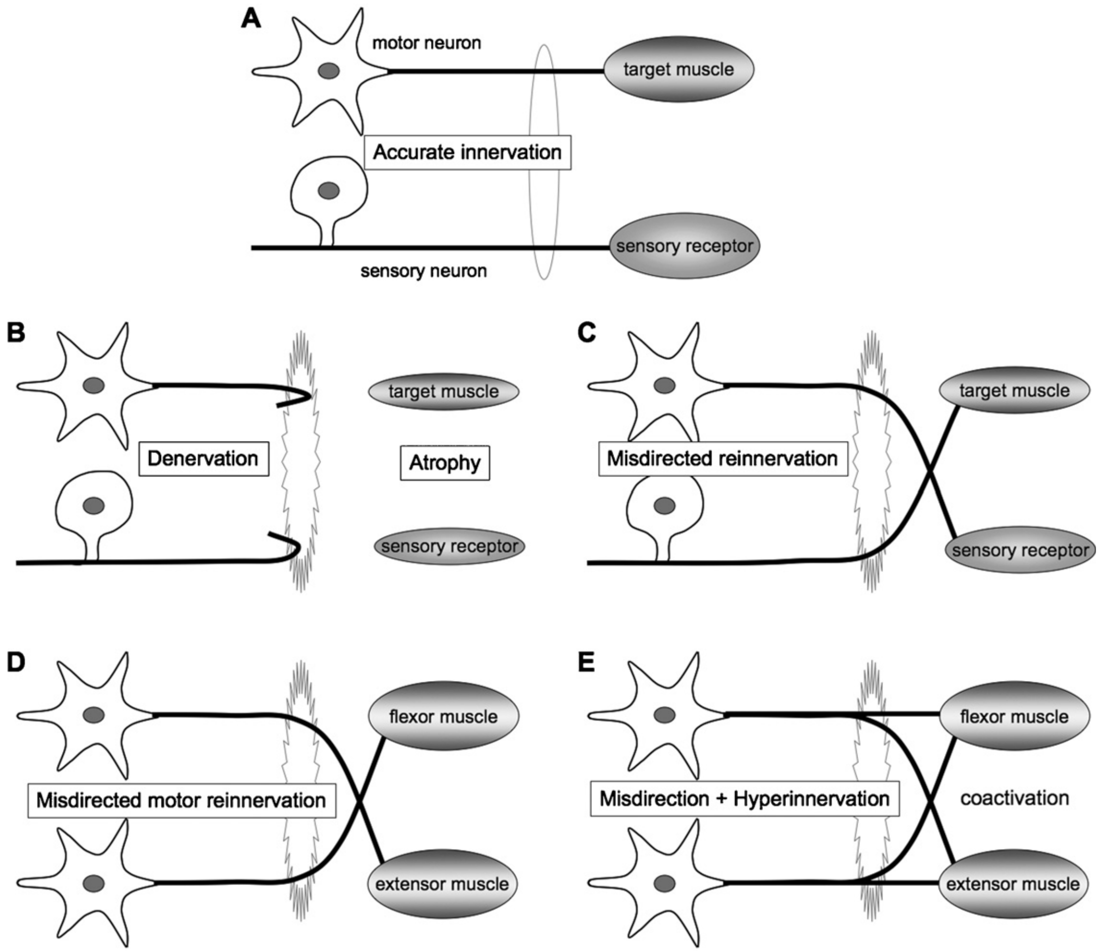

Fig. 1. Problems derived from the lack of specificity of peripheral nerve regeneration, which disrupt the normal accurate innervation of motor and sensory targets (A). (B) Abortive regeneration leads to chronic target denervation and atrophy. (C) Misdirected regeneration of axons to inappropriate targets. (D) Misdirected regeneration of axons to functionally inappropriate muscle targets. (E) Hyperinnervation of muscle targets by regenerating axons of several different neurons.

neuronal reaction is stronger after peripheral than central injuries, in which regeneration is poor and limited (Rossi et al., 2007). The increased intrinsic growth capacity of injured peripheral neurons is manifested experimentally by the conditioning lesion paradigm (McQuarrie et al., 1977). Axotomy of a peripheral neuron previous to the test lesion, "primes" the neuron, switches it on to a regenerative state and, thus, it will regenerate faster after receiving the second injury. Since the effect may require time for gene transcription (Smith and Skene, 1997), the conditioning lesion is effective if applied from 2 to 14 days before the test lesion.

Signals responsible for the initiation and maintenance of the regenerative neuronal response include a variety of mechanisms acting at sequential time phases (Fig. 2). The burst of action potentials initiated at the injury site and the disruption of the axonal transport are key points to trigger the growth capacity of neurons after axotomy (for reviews see Abe and Cavalli, 2008; Hanz and Fainzilber, 2006; Rishal and Fainzilber, 2010). At the lesion site, entrance of extracellular sodium and calcium to the injured axoplasm triggers action potentials that will be the first signals to warn the soma of the axonal injury and will provoke chromatolytic changes in the cell body (Mandolesi et al., 2004) mediated by rapid elevation of intracellular calcium and cAMP. On the other hand, the injury also disrupts the retrograde transport flow of signals from normal innervated targets, providing negative signals that inform the soma of the disconnection. Therefore, the reconnection has to be linked to recovery of the lost signals to

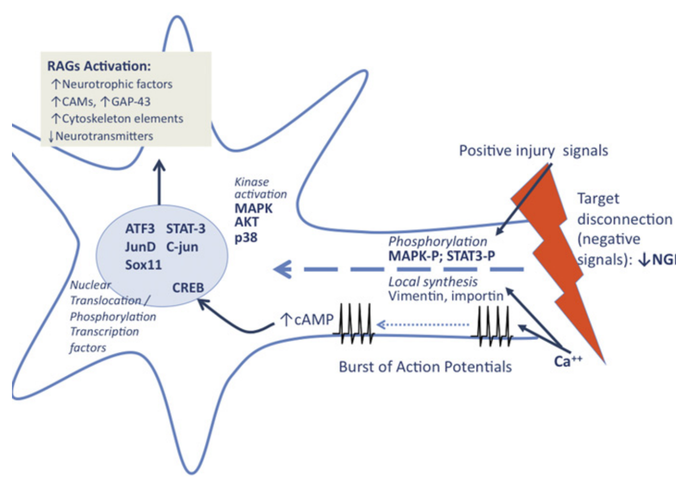

Fig. 2. Signals from the injury site arrive to the soma and switch the neuron to a pro-regenerative state, activating a set of transcription factors. Three main signaling mechanisms contribute: the burst of action potentials generated by the lesion, the interruption of the normal retrograde transport of trophic molecules provided by the end organs, and the retrograde transport of phosphorylated proteins, such as MAPK, which are positive injury signals. All these factors activate regeneration associated genes (RAGs), inducing the increased production of GAP43 and BDNF among other factors.

allow proper synaptic development. For example, transport of endogenous nerve growth factor (NGF) decreases after injury (Raivich et al., 1991) and it is considered an important negative signal to inform the neuron that its axon has been disconnected from its target. For this reason, artificial application of NGF to axotomized sensory neurons interferes with the axotomy-like changes observed in the cell body and delays axonal regrowth (Gold, 1997). Finally, there are also positive signals arriving to the soma by retrograde transport. Activated proteins, termed "positive injury signals", are endogenous axoplasmic proteins that undergo post-translational modifications at the lesion site upon axotomy, and then incorporate to the retrograde transport system for trafficking to the cell body. Several transcription factors have also been identified in peripheral axons and may participate in positive retrograde signaling (Agthong et al., 2006; Lee et al., 2004; Lindwall and Kanje, 2005). This retrograde transport involves the local synthesis of carrier proteins, including importins and vimentin, which interact with the retrograde transport motor dynein (Hanz et al., 2003). The disruption of the axon exposes the axoplasm to factors released by Schwann cells and inflammatory cells. Therefore, exposure to some trophic molecules from the injured environment, like ciliary neurotrophic factor (CNTF), neurotrophin 3 (NT3) and fibroblast growth factor (FGF) (Funakoshi et al., 1993; Kirsch et al., 2003) can induce specific responses in the neurons. For example, CNTF contacts with the tip of the injured axons and promotes phosphorylation of STAT3, which will be retrogradely transported to the soma triggering specific changes (Kirsch et al., 2003).

The different signals induced by the axonal injury lead to changes at the neuronal body. Among them, increase of intraneuronal cAMP after injury seems crucial to guarantee successful regeneration of the neurons even in non-permissive environments. cAMP can regulate axon attraction or repulsion by guidance cues from the environment (Domeniconi and Filbin, 2005; Filbin, 2003). For example, low levels of cAMP convert the attractive effects of NGF and brain derived neurotrophic factor (BDNF) into repulsion. In the case of NT3, levels of cGMP but not cAMP, regulate its responsiveness (Song et al., 1997). On the other hand, the decrease in cAMP levels in mature neurons when compared to embryonic or postnatal cells, explains the lower capacity of mature neurons to regenerate (Cai et al., 2001). Downstream effectors of cAMP are polyamines, with a strong stimulatory effect on neurite outgrowth both in vitro and in vivo (Wong and Mattox, 1991). Another downstream effector transducing the neuronal changes due to cAMP is the mammalian sterile 20-like kinase-3b (Mst3b), a serine protein-kinase which is phosphorylated by PKA (Zhou et al., 2000). The in vivo inactivation of Mst3b reduces the density of regenerating sensory axons a few days after axotomy (Lorber et al., 2009).

A key point in the changes that neurons suffer after injury is the activation of mitogen associated protein kinase cascades. Some activated intracellular signaling enzymes, such as MAPK, JNK and ERK are known to play an important role in triggering neurite outgrowth (Lindwall and Kanje, 2005; Perlson et al., 2005), whereas other kinases, as AKT and p38, have been recently described to exert early signaling responses (Michaelevski et al., 2010), although their role is still controversial. In parallel, there is nuclear localization and phosphorylation of different transcription factors, such as c-Jun, junD, ATF3, Sox11 and STAT3 (Fujitani et al., 2004; Jankowski et al., 2009; Mason et al., 2011; Nadeau et al., 2005; Schwaiger et al., 2000). Activation of these transcription factors will lead to changes in gene expression of the injured and regenerating neurons, with an increase of actin, growth associated tubulin isotypes and growth associated protein GAP-43, and a decrease of neurofilament proteins (Hoffman and Cleveland, 1988; Hoffman et al., 1987; Skene and Willard, 1981), ion channels and proteins involved in the neurotransmission machinery (Costigan et al., 2002; Jankowski et al., 2009; Mason et al., 2003; Tsujino et al., 2000).

Another possible regulator of the JAK/STAT pathway recently described that promotes axon regeneration is mTOR. Activation of the mTOR pathway is also essential for the growth capacity of peripheral neurons after injury. Blockade of the upstream inhibitors of mTOR, TSC1/TSC2 complex, mimics the conditioning lesion effect (Abe et al., 2010; Hay and Sonenberg, 2004; Park et al., 2008). In contrast, mTOR is not activated in central neurons, a fact that may contribute to their reduced regenerative response, since forced activation of mTOR in retinal ganglion cells or corticospinal neurons promotes their axonal growth (Liu et al., 2010; Park et al., 2008). The control of protein synthesis by mTOR is important in the ability of neurons to regenerate, however its expression after injury seems to be transient and it returns to baseline levels 3–4 days after injury, suggesting that a prolonged expression of mTOR might enhance axonal regeneration. Indeed, mTOR over-expression increases the levels of GAP43 after injury, but a chronic over-expression of mTOR can affect the accuracy of target reinnervation (Abe et al., 2010). Interestingly, another upstream inhibitor of mTOR, PTEN, seems to play an important role in the intrinsic capacity of peripheral neurons to regenerate, since its inhibition promotes axon growth independently of mTOR activation (Christie et al., 2010).

Some of the genes over-expressed after injury by neurons are related to increased production of trophic factors, that can act in an autocrine or a paracrine manner. After axotomy, both motor and sensory neurons increase the expression of BDNF (Kashiba and Senba, 1999; Kobayashi et al., 1996) and FGF-2 (Grothe and Unsicker, 1992; Huber et al., 1997; Madiai et al., 2003), NGF is only increased in sensory neurons (Gu et al., 1997; Shen et al., 1999), and the expression of NT-3 does not change (Shen et al., 1999). The expression of different receptors for neurotrophic factors is also differentially regulated after axotomy (see below).

Another relevant change in the neuronal phenotype after axotomy is the downregulation of neurotransmitters and transmitter-related proteins. However, there are also marked changes in the expression of neuropeptides in the axotomized neurons, which vary in different types of neurons and contribute to neural signaling after injury (Hökfelt et al., 2000). Thus, motor neurons characteristically show a rapid and long-lasting increase in calcitonin gene-related peptide (CGRP) immunoreactivity (Borke et al., 1993), whereas CGRP, as well as substance P (SP) are downregulated after injury in small and medium size primary sensory neurons (Dumoulin et al., 1991; Noguchi et al., 1990; Shadiack et al., 2001; Villar et al., 1991), due to interruption of target-derived NGF to the neuronal body (Eriksson et al., 1997). However, CGRP and SP appear to increase at the regenerating front, where they are synthesized at the tip of the growing axons (Li et al., 2004; Toth et al., 2009; Zheng et al., 2008). Blocking CGRP synthesis at the axonal level with siRNA and blocking its receptor on Schwann cells were found to impair nerve regeneration and Schwann cell recruitment during neurite growth (Raivich et al., 1992; Toth et al., 2009). Other peptides such as vasoactive intestinal polypeptide, galanin and neuropeptide Y, which are normally expressed at low levels in sensory neurons, are markedly increased in those neurons (Villar et al., 1989; Villar et al., 1991; Wakisaka et al., 1991), and transiently upregulated also in motor neurons (Zhang et al., 1993) after axotomy. These changes seem to be involved in regenerative and/or compensatory processes following nerve damage.

3. Biology of axonal elongation: the growth cone

The regeneration of a sectioned axon involves the transformation of a stable axonal segment into a highly motile tip, called growth cone, that senses the surrounding environment and leads the elongation of the regenerating axon. It is important that growth cones elongate following the endoneurial tubes of the distal nerve to eventually reinnervate their peripheral target organs. When growth cones do not reach the distal stump they may sprout within the proximal stump forming a neuroma. In the distal stump, denervated Schwann cells proliferate on the basal membrane of the endoneurial tubes and form columns, the so-called bands of Büngner, over which growth cones advance (Bunge et al., 1982). Each regenerating axon may give rise initially to over 10 axonal sprouts (Witzel et al., 2005), but the number of branches decreases with time in the distal segment, as sprouts that do not make peripheral connections undergo atrophy and eventually disappear, whereas axons reinnervating target organs mature and enlarge in size. Nevertheless, even several months after injury, regenerating neurons may support multiple sprouts, branching from the level of the lesion, in the distal nerve stump. This fact explains the higher counts of nerve fibers in the distal stump compared to the real number of proximal fibers that regenerate across the lesion (Horch and Lisney, 1981; Jenq and Coggeshall, 1985; Mackinnon et al., 1991).

The orientation of the advancing tip is guided by gradients of neurotrophic and neurotropic factors produced mainly by non-neuronal cells (Fig. 3). This process is also known as chemotaxis. The molecular cues that the regenerating axon will find at the lesion site and into the distal nerve stump can have both chemoattractive and chemorepulsive properties, and can be diffusible or membrane-bound. Indeed, the different signals may create a permissive environment for axonal growth or constitute a molecular barrier that impairs axonal elongation (for reviews see Dickson, 2002; Mueller, 1999; Song and Poo, 1999; Tessier-Lavigne and Goodman, 1996; Wen and Zheng, 2006).

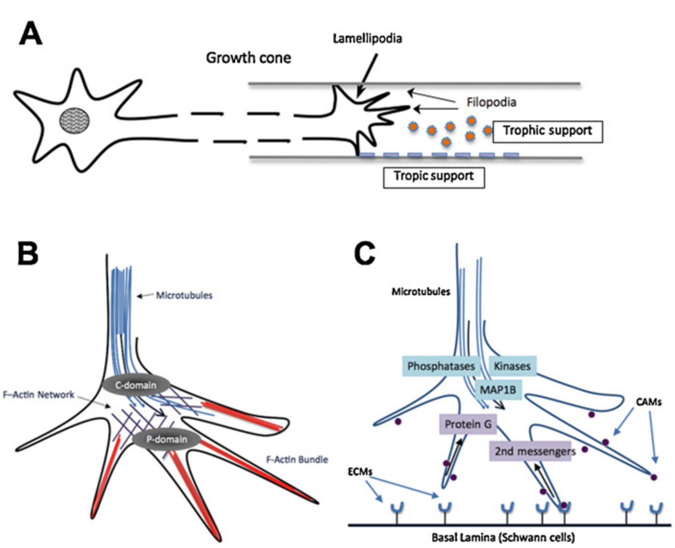

Fig. 3. (A) Neurotrophic and neurotropic factors are needed during peripheral nerve regeneration and determine the response of the growth cone, which extends lamellipodia and filopodia depending on the different attractive and repulsive gradients present in the extracellular environment. Neurotrophic factors are diffusible molecules like NGF, GDNF, BDNF or NT3, while neurotropic factors are substrate-bound cues such as laminin, fibronectin and collagen or cell adhesion molecules, that maintain hemophilic interactions. (B) Filopodia and lamellipodia are the mobile path-finder parts of the growth cone. When filopodia encounter a permissive substrate, growth cone receptors bind to ECM molecules. These receptors are coupled to F-actin fibers that re-organize the cytoskeleton and lead growth cone protrusion. F-actin polymerization pushes the membrane forward at the protrusion domain, while microtubules already stabilized at the central domain localize and polymerize depending on actin network. (C) The extracellular matrix influences the state of the regenerating neurons and determines the response of the growth cone. Interactions between substrates and their receptors change the intracellular levels of second messengers, as for example cAMP, in the growth cone. Interactions between ligands and CAMs activate small GTPases that play a fundamental role in growth cone motility and cytoskeleton remodeling.

The growth cone advances by three main steps: protrusion, engorgement and consolidation (Mortimer et al., 2008). The growth cone shows three functionally specific regions: a central domain (C-domain), a transition zone (T-zone) and a peripheral domain (P-domain) (Bouquet and Nothias, 2007). The T-zone is placed between the microtubule-rich C-domain and the actin-rich P-domain. Actin polymerization triggers the protrusion of growth cone membrane and this facilitates the delivery of new microtubule segments at the P-domain, as well as their stabilization along actin bundles. This allows the displacement of the C-domain forward (engorgement) and the further consolidation of the nascent axon. Addition of soluble tubulin into the distal ends of microtubules that enter the C-domain leads to microtubule polymerization, whereas high concentrations of myosin at the T-zone lead to the contraction of the actin network, inducing formation of an actin-filament arc. The most distal part of the growth cone, the P-domain bears lamellipodia, membranous protrusions from which extend several expansions called filopodia, mainly formed by actin filament bundles. Equilibrium between actin polymerization and depolymerization generates constant protrusion forces, thus allowing filopodia to act as the "tentacles" of the growth cone, pulling and pushing to explore the microenvironment.

Cytoskeleton proteins of regenerating peripheral axons undergo qualitative and quantitative changes similar to those of growing axons during development (Hoffman and Cleveland, 1988). Transection of motor and sensory fibers increases the levels of tubulin isoforms, actin and peripherin, and decreases the levels of neurofilaments that regulate axon caliber. The enhanced metabolic requirements of the growing axons are sustained by an increased synthesis of different components at the neuron body that will be anterogradely transported to the fiber tip (Hoffman and Lasek, 1980; McQuarrie et al., 1989), although there is also local synthesis and degradation of proteins in the axon (Verma et al., 2005; Willis and Twiss, 2005). During the final consolidation phase, new microtubules are assembled in the C-domain, probably mediated by the activity of microtubule- and actin-associated proteins.

Factors that may be involved in the control of microtubule assembly in growth cones include calcium, cyclic nucleotides, post-translational modification of tubulin and microtubule-associated proteins (MAPs) (Gordon-Weeks and Mansfield, 1992). When the growth cone has moved on and microtubules have become incorporated into the neurite cytoskeleton, post-translational modifications of tubulin increase the stability of microtubules to depolymerization (Schulze et al., 1987). The stabilization of microtubules in a permanent cytoskeleton is also dependent on a set of structural proteins that can be classified as MAP (MAP1–MAP5) and tau proteins; both classes induce polymerization of tubulin and remain bound to the newly formed microtubules. Tau proteins bind to several tubulin molecules simultaneously, thereby enhancing tubulin polymerization. MAPs act similarly, but they can cross-link the microtubule to other cell components. All known MAPs are highly phosphorylated proteins, and thus substrates for phosphorylating enzymes and GAPs, particularly GAP-43. GAP-43 expression is closely related with periods of active growth cone function, and its levels are dramatically increased within regenerating nerves after axonal injury (Benowitz and Lewis, 1983; Skene, 1989).

The contribution of cytoskeletal proteins to axonal regeneration is crucial (Lewis and Bridgman, 1992). In fact, despite the diversity of signaling pathways that can lead to or inhibit axonal growth, ultimately any factor that alters the growth capacity of neurons implies changes in the cytoskeleton components (for reviews see Lowery and Van Vactor, 2009; Tang, 2003). The small GTPases of the Rho family (being RhoA, Rac1and Cdc42 the most well-known) integrate the upstream signaling cues for the downstream cytoskeleton rearrangements in a spatial-specific manner, reproducing at the cytoplasmatic level, the extracellular environmental changes. Receptor complexes present at the growth cone can recruit GTPases, which are needed to control the contractile capability dependent on actin-myosin interactions (Luo, 2002). Although numerous signaling transduction molecules can convey guidance information, the Rho-family and their effectors, such as Rho-associated kinases (ROCKs) among others, are probably the key points in the intracellular signaling cascades that promote regeneration (Tang, 2003). Activation and regulation of these molecules and their effectors is complex, and different GTPases can be activated by the same guidance cues. On the other hand, different trophic molecules, as for example NGF, and the extracellular matrix molecule laminin, regulate neurite outgrowth acting coordinately on the same GTPase (Rankin et al., 2008). Nowadays, it is considered that the spatial-temporal localization and activation of small GTPases by different trophic molecules is essential for their functional outcome (Pertz et al., 2008).

Guidance cues of the external environment also play a main role in the elongation of the growth cone. Guidance cues interact with specific receptors on the surface of the growth cone, and trigger a cascade of cytoplasmic events that eventually lead to the cytoskeletal rearrangement associated with oriented neurite extension. For example, cell adhesion molecules (CAMs) mediate the adhesion of filopodia to tropic molecules of the extracellular matrix (Letourneau and Shattuck, 1989). Moreover, lipid rafts (see below) present at the growth cone amplify guidance cue signals (Fujitani et al., 2005; Guirland et al., 2004); neurotrophins, for instance, can exert their chemoattractive gradient acting through these microdomains.

During development, axons use at least two types of cues as they grow along the spinal nerves and into peripheral nerves. First, they respond with differential attraction between different types of axons, an effect mediated by changes in expression of several CAMs. Second, axons can also be guided by molecular cues, either attractive or repulsive, coming from the local environment of the limb. Both motor and sensory axons are specified with respect to their peripheral targets during development, thus their axons have to respond to a set of guidance signals as they travel through the spinal plexus, the peripheral nerve trunk and successive branching points to an adequate target. Motoneurons project axons out of the central nervous system to muscle targets located at the periphery. Distinct pools of motoneurons, corresponding to fast/slow and flexor/extensor subclasses seem to be specified at embryonic stages (Landmesser, 2001). During development, motoneurons are able to grow their axons to appropriate muscles, in response to sets of guidance cues, presumably inhibitory, provided by non-myogenic cells within the limb. The decision of a motor axon to project to dorsal or ventral muscles (functionally antagonists) in the limb is controlled in part by signaling through several ephrins and their receptors at the plexus level (Schneider and Granato, 2003). Muscular sensory axons grow slightly later than motor axons and they become adjacent and fasciculate along motor axons innervating the same muscle. In contrast, developing cutaneous sensory axons bundle together, and at short distance from the dorsal root ganglia (DRG), axons that project along individual cutaneous nerves are already restricted in their position (Honig et al., 1998). The achieved pattern of peripheral projections has to be maintained throughout life for an adequate functional control of skeletal muscles, as well as for the accurate localization and perception of stimuli affecting cutaneous or muscular sensory receptors. It has been reported that members of the three main families of guidance molecules (Netrins, Semaphorins and Slits) play key roles as molecular barriers or promoters for sensory axon growth during development (Del Río et al., 2004). For motor axons, molecules such as T-Cadherin, HGF/SF as well as Ephrins have been involved during their development (Gavazzi, 2001; Krull and Koblar, 2000). However, despite the similarities of axonal growth during development and post-injury regeneration, the regulation and participation of the former molecules during peripheral nerve regeneration and selective sensory and motor axon outgrowth remain unclear. In addition, the growth cone integrates the information of several extracellular signals and convergent intracellular pathways. Indeed, numerous in vitro growth cone turning assays indicated that intracellular kinases (e.g., ERK1/2, PI3K, GSK3β (Liu and Snider, 2001) and signaling cascades activated by axonal guidance receptors are modulated by cyclic nucleotides cAMP and cGMP (Song and Poo, 1999). For example, elevated cGMP levels may counteract the inhibitory effects of Sema3A on sensory neurons (Schmidt et al., 2002).

Several studies have reported that motor and sensory fibers respond differently after nerve lesions (Hoke et al., 2006; Lago et al., 2007). CAMs differentially expressed by sensory or motor Schwann cells may play a role for preferential guidance. Preferential reinnervation is thought to be due to collaboration between the regenerating axons and the appropriate pathway (Brushart, 1993; Madison et al., 1996), although attempts to promote specific axon guidance using biochemical methods had modest success until now. Moreover, a detailed study of factors able to induce specific regeneration of motor and sensory axons is currently not available.

4. Regeneration along the distal nerve stump

In order to achieve successful regeneration, neurons have to survive and to activate the re-growing program that will lead to axonal elongation and associated plastic changes of their network. This program is influenced by interactions between the neurons and the neighboring glial cells, mediated by cell- and substrate-adhesion molecules and their receptors, and by several trophic factors secreted into the extracellular space. A number of key players in this regenerative process have been identified, but the relationship between molecular events, particularly the triggering of gene expression and the corresponding cascade of signaling pathways, is still incompletely understood.

The environment that the axons find distal to the lesion site is crucial for their regeneration and thus, it is important to create a favorable milieu for axonal outgrowth. An intact peripheral nerve does not support axonal growth as effectively as a degenerated nerve, due to the presence of inhibitory factors for regeneration (Mueller, 1999; Tang, 2003), such as chondroitin sulfate proteoglycans (CSPGs) of the extracellular matrix and myelin-associated inhibitors of regeneration (Mukhopadhyay et al., 1994). These molecules, upregulated after nerve injury, show neurite-inhibitory activity (Braunewell et al., 1995; Shen et al., 1998; Zuo et al., 1998) and contribute in delaying axonal growth following the lesion. However, the distal nerve stump undergoes Wallerian degeneration, a process in which activation of Schwann cells and recruitment of macrophages (Stoll and Muller, 1999) lead to the lysis and elimination of the distal part of axons and myelin debris, with consequent clearance of inhibitory factors. Proliferation of Schwann cells is not only important to initiate the phagocytosis of myelin sheaths and to trigger the recruitment of infiltrating macrophages, but also to create a permissive environment for axonal growth. De-differentiated Schwann cells covered by the basal lamina constitute a permissive substrate along which axons will elongate (Fig. 4). Interactions between the axons, the Schwann cells and the basal lamina are maintained by different adhesion molecules like cadherins, immunoglobulin cell-adhesion molecules (Ig CAM) and integrins (Ide, 1996; Huber et al., 2003).

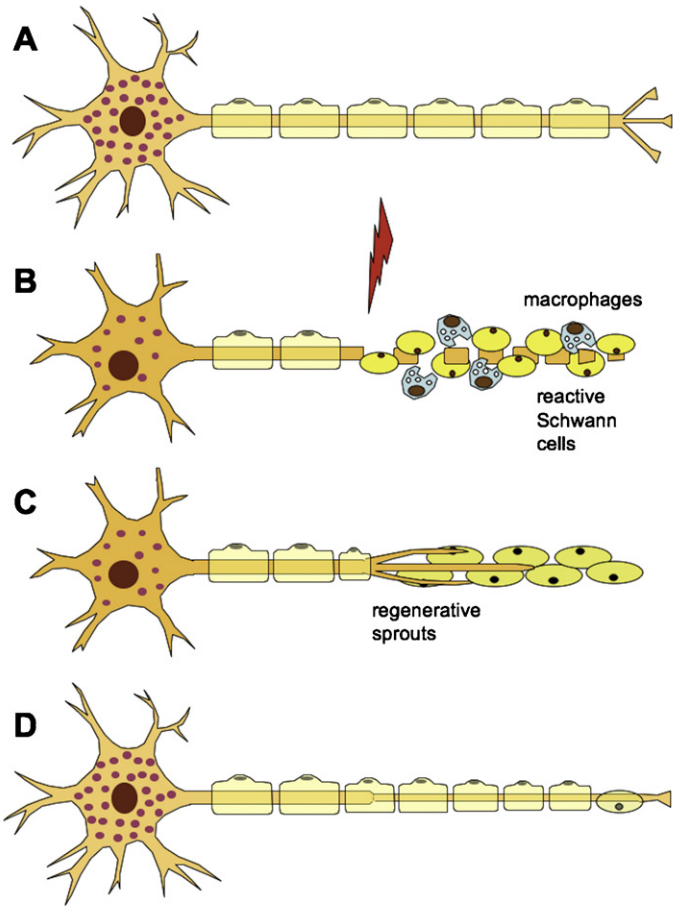

Fig. 4. Degeneration and regeneration after peripheral nerve injury. (A) Normal neuron and nerve fiber. (B) Wallerian degeneration. The axotomy results in fragmentation of the distal axon and myelin sheaths. Schwann cells proliferate and macrophages invade the distal nerve segment and phagocytose degrading materials. (C) Schwann cells in the distal segment line up in bands of Bungner. Axonal sprouts advance embedded in the Schwann cells and attracted by gradients of neurotrophic factors. (D) Axonal reconnection with end organs and maturation and remyelination of the nerve fiber.

Regenerating axons have to cross the injury site prior to reaching the distal stump. After axonotmesis, as produced by a nerve crush, there is only disruption of the axons but the perineurium and the endoneurial tubules are preserved, so the axons usually reach the distal stump with no major delays. However, after complete nerve transection, when the continuity of the connective tissue is lost and elongation towards the distal nerve is not allowed, a neuroma usually forms at the severed nerve end, which is frequently associated with significant pain and dysesthesia. Regenerative axonal sprouts continue to grow blindly within a dense collagenous and fibroblastic stroma in the neuroma. In contrast, if the continuity of the nerve is re-established by means of a direct suture of the stumps, axons can readily elongate along the distal degenerating nerve, although they have to cross the scar tissue at the rejoined site that imposes some delay. When primary suture is not possible, a bridge constituted by an autologous nerve graft is an adequate alternative (Lundborg et al., 2004). An important impediment for a good recovery after nerve transection is the loss of the normal fascicular architecture of the peripheral nerve after the surgical repair, with axons growing dispersed in numerous mini-fascicles (Lago and Navarro, 2006). Consequently, the topographical peripheral projections of motor and sensory neurons are distorted and functional recovery is hampered (Gramsbergen et al., 2000; Valero-Cabre and Navarro, 2002; Valero-Cabre et al., 2004). The type of nerve graft used has been related to the level of recovery; motor nerve grafts have been found to allow more robust regeneration and better functional outcome than sensory nerve grafts (Moradzadeh et al., 2008; Nichols et al., 2004), possibly related to the larger endoneurial tubes of motor nerves. Another study provided evidence that regeneration of motor axons was promoted at earlier times by a motor nerve graft (from ventral spinal root), whereas reinnervation of sensory pathways was slightly improved in the presence of a sensory graft (from dorsal root) (Lago et al., 2007). Findings by Redett et al. (2005) showed that motoneurons maintain more regenerating collaterals in cutaneous than in muscular distal nerves. Such difference could be due either to increased collateral formation in cutaneous nerve or to increased collateral pruning in muscle nerve. In either instance, they indicate that motor and sensory pathways have distinct functional identities that regulate the regenerative sprouting of axons. Nevertheless, the potential differential contribution of Schwann cells in either motor or sensory nerves has been also suggested.

5. Changes in Schwann cells after nerve lesions

Schwann cells are derived from the neural crest and their precursors are able to proliferate until they differentiate to mature myelinating and non-myelinating phenotypes. Interestingly, mature Schwann cells have the remarkable capacity to reverse their phenotype and dedifferentiate when they lose contact with axons. Therefore, after a peripheral nerve injury, the molecular markers characteristic of myelinating and non-myelinating Schwann cells are downregulated (Fig. 5). In particular, expression of myelin markers decreases dramatically as a consequence of axonal degeneration distal to the injury site, whereas markers of immature and non-myelinating Schwann cells are re-acquired (Scherer et al., 1994; Scherer and Salzer, 1996). The loss of the transcription factor Krox-20 after nerve section or axonotmesis (crush) probably allows the cells to resume their constitutive c-Jun expression, which will help the glial cells to de-differentiate and proliferate (Parkinson et al., 2008). The phenotype of reactive Schwann cells resembles the one of immature Schwann cells and they form a permissive substrate for regeneration. When these

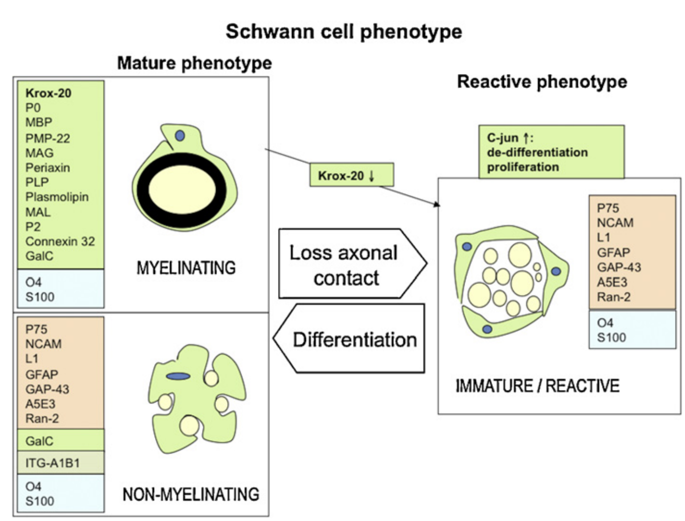

Fig. 5. Immature Schwann cells are highly proliferative cells until they differentiate into the two mature phenotypes: myelinating and non-myelinating, depending upon the type of axon they ensheath. After injury, due to the loss of contact with the axons, myelinating Schwann cells lose the transcription factor Krox-20, down-regulate molecules related to myelin and recover the constitutive c-Jun expression, that will allow the cells to return to a proliferative-regenerating phenotype similar to the immature state.

cells regain contact with the axons, they re-differentiate again (Jessen and Mirsky, 2008).

Between days 1 and 5 after injury, Schwann cells start proliferating and their peak of activation occurs around day 3 and then decreases during the following weeks (Bradley and Asbury, 1970). This proliferation plays a key role during Wallerian degeneration since, in coordination with macrophages, Schwann cells initiate degradation of myelin debries. A second phase of proliferation occurs during the regenerative process. As they proliferate within the basement membrane of the endoneurial tubules, Schwann cells form distinct rows called bands of Büngner, which create a permissive pathway to regenerating axons.

The loss of contact with the axolema induces the Schwann cell to express on its plasma membrane several receptors for neurotrophic factors and to produce and release several neurotrophic and neurotropic molecules. Denervated Schwann cells over-express fibronectin, laminin, tenascin and some proteoglycans, which create a substrate for axonal elongation. They also increase expression of several neurotrophic factors, such as NGF, BDNF, NT-4, glial cell-derived neurotrophic factor (GDNF) and insulin-like growth factor-1 (IGF-1). Interestingly, the time course of expression after the injury is different for each factor (Gillen et al., 1997). In contrast, Schwann cells down-regulate the production of myelin proteins, as well as some other trophic factor, as for example CNTF (Rabinovsky et al., 1992).

As regenerating axons enter the distal portion of the nerve, they are guided haptotactically along the Schwann cell substratum by binding of trophic factors to their own receptors. The close apposition of regenerating axons and Schwann cells facilitates the receptor-mediated, intercellular transfer of neurotrophic factors. Once Schwann cells regain the contact with axons, the expression of neurotrophic factors and their receptors is suppressed. This mechanism creates a dynamic gradient where the highly activated Schwann cells are located distally into the degenerated nerve stump. This gradient of substrate-adsorbed neurotrophic factors would help to maintain the proper directionality of axonal regeneration. When the neuronal axons reinnervate target organs, Schwann cells return to a quiescent state (Taniuchi et al., 1988). The regenerating axons that are advancing into the endoneurial tubes give signals to the Schwann cells making them turn to a mature phenotype. Interestingly, the regenerating axons originated by myelinated neurons instruct the Schwann cells to form myelin, while unmyelinated fibers remain unmyelinated independently of the origin of the Schwann cells they encounter in the distal pathway (Politis and Spencer, 1981).

Besides the myelinating or unmyelinating phenotype, Schwann cells from motor and sensory nerve branches appear to express different molecular markers that may contribute to the capacity of some axons to specifically regenerate towards appropriate pathways and reinnervate corresponding target organs. It was shown that Schwann cells associated to motor but not to sensory neurons differentially express the L2/HNK1 carbohydrate epitope (Martini et al., 1992). Interestingly, after injury it seems that the expression of L2/HNK1 by motor associated Schwann cells is lost, but if these cells recover contact with regenerating motor axons, they show a stronger tendency to express L2/HNK1 than Schwann cells formerly associated with sensory axons, although they also enter in contact with motor axons (Martini et al., 1994). On the other hand, NCAM-positive cells have been localized exclusively in sensory nerve fascicles, and they have been identified as non-myelin forming Schwann cells of sensory unmyelinated fibers (Saito et al., 2005). Both markers appear distinctly by about 3 weeks postnatally, suggesting that they play a role during development of nerve fasciculation. However, HNK-1 and NCAM immunoreactivity is lost within weeks/days after nerve transection in motor and sensory fascicles, respectively (Saito et al., 2005), although it may reappear after recontact with the appropriate axons (Martini et al., 1994). This fact suggests that Schwann cells dedifferentiate after injury and lose the expression of specific markers decisive for their phenotype, but they retain some of their acquired properties.

Regarding the neurotrophic profile, Hoke et al. (2006) observed that Schwann cells of sensory and motor (ventral roots) nerves exhibit differing growth factor profiles at baseline and, interestingly, respond differently during denervation and when reinnervated by cutaneous or motor axons. mRNAs for NGF, BDNF, vascular endothelial growth factor (VEGF), IGF-1 and hepatocyte growth factor were expressed vigorously by denervated and reinnervated cutaneous nerves but minimally by ventral roots, whereas mRNAs for GDNF and pleiotrophin were increased to a greater degree in motor than in cutaneous nerves. Such differences were maintained during Wallerian degeneration, although they tended to decline with time when reinnervation was induced by the wrong type of axons (Hoke et al., 2006). Immunohistochemical analyses of the expression of neurotrophins have shown that Schwann cells around intact sensory axons of the dorsal root and cutaneous saphenous nerve exhibit a higher expression of NGF, BDNF and NT-3 than those around intact motor axons. In contrast with intact nerves, distal segments of transected saphenous nerve and muscular femoral nerve displayed much higher intensity of immunostaining for neurotrophins, but at similar levels when compared with each other (Dubovy, 2004). These results indicate that upregulation of neurotrophins in the reactive Schwann cells of the injured peripheral nerve provides similar conditions for the growth of both sensory and motor axons, suggesting that trophic factors may not contribute significantly to the sorting of different types of axons during nerve regeneration. Further investigations addressed to enhance the selective phenotype of motor or sensory Schwann cells, by engineering them to overexpress certain neurotrophic factors or membrane adhesion molecules, might help to elucidate the above discrepancies and promote selective axonal regeneration.

The key role of active Schwann cells in axonal regeneration is evidenced by the poor functional recovery achieved when using acellular nerve grafts to repair nerve resections (Hall, 1986). Similarly, the slow decrease in the capacity of Schwann cells to maintain an active pro-regenerative phenotype with time explains the limited capacity of chronically denervated nerves to sustain axon regeneration, and reveals the importance of early nerve repair and strategies to accelerate regeneration (Gordon et al., 2003). After severe injuries of proximal nerves, that require long periods of time for regeneration, the poor motor recovery reached has been related to the decreased capacity of Schwann cells to provide support for regeneration (Sulaiman and Gordon, 2000), together with the regression growth state of the chronically axotomized neuron (Fu and Gordon, 1995a) and the atrophy of the denervated muscle (Fu and Gordon, 1995b; Gordon et al., 2011; Ma et al., 2011). Interestingly, a recent work points out the importance of terminal Schwann cells of the motor endplate in failed reinnervation of chronically denervated muscles. After a window of opportunity, motor axons that reach the target organ fail to reinnervate the endplate. In contrast, sensory axons are able to reinnervate the skin in a similar degree both at early or later stages. Schwann cells of the motor endplate could become non-permissive for regeneration after long periods of denervation and prevent functional synapses of the motor axons with the muscle (Ma et al., 2011).

6. Neurotrophic factors: effects on different neurons after injury

Large quantity of studies depicts the important role of neurotrophic factors in determining neuronal survival both during development and after injury. Under physiological conditions, non-neuronal cells synthesize and secrete several trophic factors, needed for maintenance of the homeostatic state of intact neurons. Schwann cells and several peripheral tissues (muscle, skin, etc.) have a basal expression of neurotrophic factors, which play biological functions in promoting survival and maintenance of sensory, motor and autonomic peripheral neurons (Gordon, 2010; Lewin and Barde, 1996; Raivich and Kreutzberg, 1994). After nerve injury the expression of those neurotrophic factors increases, in an attempt to support the survival and growth of regenerating axons and reinnervation of denervated target organs. The changes in the distal nerve follow different patterns for different trophic factors; thus, there is an initial up-regulation of some of them, like NGF, BDNF and GDNF, whereas others, as NT-3 and CNTF, are down-regulated. Their levels return to normal with regeneration, but in case of chronical denervation the distal nerve stump may continue to express neurotrophic factors for at least 6 months following injury (Michalski et al., 2008), although other studies point the failure of Schwann cells to maintain high levels of trophic factors after 2 months of chronic denervation (Gordon, 2009).

6.1. Neurotrophins

The neurotrophin family includes NGF, BDNF, and neurotrophins NT-3, and NT-4/5. Among them, NGF is the most studied one. It is known to act specifically on a subpopulation of small primary sensory and on sympathetic neurons (Levi-Montalcini, 1987). It seems to be expressed at similar levels in intact cutaneous and motor nerves, but after injury there is an up-regulation in denervated sensory dorsal roots to a larger extent than in motor ventral roots (Hoke et al., 2001). In adult animals, NGF is needed for collateral sprouting of nociceptive and sympathetic axons into denervated skin, but does not affect large sensory axons (Diamond et al., 1987; Gloster and Diamond, 1992) In contrast, application of NGF after axotomy delays the onset of regeneration (Gold, 1997), probably by reducing the neuronal body response to injury (Mohiuddin et al., 1999), although without compromising the rate of subsequent regeneration. On another hand, focal administration of NGF or NT-3 to adult axotomized nerves inside a blinded impermeable chamber effectively prevented sensory neuronal loss, which mainly affects small DRG neurons (Rich et al., 1987; Groves et al., 1999).

Another neurotrophin, BDNF, is differently expressed in motor and sensory intact nerves, and after injury, it can be detected also at higher concentrations in cutaneous nerves than in ventral roots (Hoke et al., 2006). The role of BDNF in nerve regeneration is controversial, although the presence of its receptor trkB seems crucial to sustain axonal regeneration (Boyd and Gordon, 2001). Deprivation of endogenous BDNF impairs axonal growth and myelination (Zhang et al., 2000), whereas local infusion of BDNF improved nerve regeneration in neural conduits (Vögelin et al., 2006). Interestingly, Boyd and Gordon (2001) described no effects on axonal regeneration when BDNF was administered acutely after a cut and suture of the sciatic nerve, although it enhanced regeneration in a dose dependent manner when applied after chronic axotomy (Boyd and Gordon, 2003).

Neurotrophin-3 was shown to be present at higher concentration in cutaneous sensory than in motor nerves in baseline conditions, but no great differences between roots were found after injury (Hoke et al., 2006). NT-3 is present in adult skeletal muscles and has a trophic role on motoneurons and primary sensory neurons innervating muscles. Its trophic action on motoneurons was shown in vitro (Braun et al., 1996), whereas in vivo it seems to exert a selective action on type 2b fast muscle fibers (Sterne et al., 1997a,b). On the other hand, other studies (Ernfors et al., 1993; Airaksinen et al., 1996) confirm its important role for the survival of proprioceptive and mechano-receptive sensory neurons. Moreover, NT-3 can be found in trigeminal, cervical and lumbar spinal ganglia (Hory-Lee et al., 1993; Zhou and Rush, 1995). It is interesting to highlight that in chicken embryos, NT-3 responsive sensory neurons extend neurites well both on laminin and fibronectin coated coverslips, whereas NGF responsive neurons grow better on laminin. Thus, differences seem to exist in substrate preference between NGF and NT-3 responsive sensory neurons (Guan et al., 2003).

Focal delivery of BDNF, NT-3 or NT-4 into fibronectin guides revealed that NT-4 preferentially improved the functional reinnervation of slow motor units, whereas BDNF and NT-3 showed less effects on regenerating motoneurons (Simon et al., 2003). In another study, exogenous administration of NT-3 was selectively beneficial for the reinnervation of 2b fast muscle fibers (Sterne et al., 1997a,b), suggesting that different growth factors might preferentially influence different type of motoneurons. The use of specific growth factors could also be useful to selectively guide axons from a mixed population of DRG neurons. By using a compartmentalized delivery of NGF and NT-3 in vitro, Lotfi et al. (2011) tried to preferentially enhance growth of nociceptive and proprioceptive subsets of sensory neurons, respectively. They found that the NGF channel attracted nociceptive axons, that elongated longer when compared to saline or NT-3 channels, whereas there were more proprioceptive fiber branches in the NT-3 channels. However, the authors failed to corroborate these findings in vivo.

The specific receptors for neurotrophins are called Trks, and are a family of protooncogene receptors, which mediate their effects on distinct populations of DRG neurons from E15 onward and in some motoneurons as well. TrkA is the high affinity receptor for NGF and it is expressed in subpopulations of primary sensory neurons, especially in the small ones (Bennett et al., 1996a,b). Interestingly the expression of this receptor is higher in visceral afferents than in cutaneous afferents (Mu et al., 1993). The high affinity receptor for BDNF and NT-4/5 is TrkB, normally found in spinal motoneurons and also in mid-size primary sensory neurons (Klein et al., 1989, 1992; Mu et al., 1993). After axotomy, motoneurons increase the expression of TrkB (Ernfors et al., 1993; Kobayashi et al., 1996; Piehl et al., 1994). TrkC, the high affinity receptor for NT-3, is present in spinal motoneurons and in a subpopulation of large diameter primary sensory neurons (Lamballe et al., 1991; Mu et al., 1993). In addition, it is important to consider that neurotrophins have also a low affinity receptor P75, whose expression increases mainly after injury. The role of this receptor is controversial as it can exert also pro-apoptotic functions. Following peripheral nerve injury, neurotrophin receptors are upregulated on the distal portion of the nerve, in denervated Schwann cells and in growth cones of regenerating axons (Raivich and Kreutzberg, 1994), although differentially regulated. Thus, reduced levels of TrkB and TrkC receptors are detected in Schwann cells of the distal injured nerve (Funakoshi et al., 1993), whereas TrkA has been found to decrease in the DRG neurons perikarya, but increase at the regenerating nerve front (Webber et al., 2008). After axotomy there is an increased transport of all the neurotrophins dependent on neurotrophin binding to the P75 and Trk receptors in sensory neurons, but only to P75 that is upregulated in motoneurons (Curtis et al., 1998).

6.2. Glial derived neurotrophic factor

GDNF belongs to the TGFα-1 superfamily and has a trophic effect on sensory, motor and autonomic neurons (Buj-Bello et al., 1995; Ebendal et al., 1995; Henderson et al., 1994a; Trupp et al., 1995; Matheson et al., 1997; Baloh et al., 1998). Two weeks after birth, some of the small-diameter non-peptidergic neurons of the

DRG (a subpopulation that can be identified as lectin IB4 positive) lose their sensitivity to NGF and become GDNF sensitive (Bennett et al., 1996a; Molliver et al., 1997). The IB4-labeled sensory neurons have an impaired intrinsic axonal regeneration capacity and higher vulnerability to axonal injury than the NGF-dependent sensory neurons, even if exposed to GDNF supply (Leclere et al., 2007). In contrast, overexpression of GDNF in motoneurons has profound positive effects on neuronal survival after axotomy (Zhao et al., 2004) and prevents axotomy-induced ChAT decrease (Henderson et al., 1994a; Yan et al., 1995).

Although it has been shown that GDNF levels are higher in cutaneous nerves than in ventral roots in intact animals (Hoke et al., 2006), this factor is similarly up-regulated both in injured ventral and dorsal roots (Hoke et al., 2001). This up-regulation of GDNF after injury triggers also the up-regulation of its receptor (GFRα-1) suggesting its important role for neurotrophic support (Höke et al., 2000). In fact, several studies showed that GDNF enhances regeneration of both motor and sensory axons. For example, sustained delivery of GDNF to the injury site in vivo by a synthetic nerve guide allowed regeneration of both sensory and motor axons over long gaps to a significantly higher number than with NGF delivery (Fine et al., 2002). In DRGs, GDNF delivery enhances the conditioning injury effect, although only at low GDNF concentrations (Mills et al., 2007). Furthermore, GDNF expression decline was associated with impaired regeneration after long-term denervation (Hoke et al., 2002), whereas GDNF applied to the proximal stump of chronically sectioned nerves increased the number of motoneurons that were able to regenerate their axons (Boyd and Gordon, 2003). GDNF has also important effects on Schwann cells, and when applied at high doses in adult rats induces proliferation of Schwann cells and the myelination of axons, even the normally unmyelinated ones (Höke et al., 2003). This fact suggests that GDNF can mediate axon-glial interactions. Schwann cell-derived GDNF is taken up by sensory and motor neurons and transported anterogradely along the axons for release from terminals, where it may act on glial cells expressing its receptor GFR.

The GDNF receptor complex is composed of the ligand binding receptors GFRα and the signal-transducing domain RET (Treanor et al., 1996; Bennett et al., 2000; Trupp et al., 1996). GDNF seems to bind preferentially GFRα-1, while GFRα-2, GFRα-3 and GFRα-4 are supposed to bind preferentially neurturin, persephin and artemin. GDNF forms a high-affinity complex with GFRα-1, and then this complex recruits Ret, induces phosphorylation of specific tyrosine residues of Ret and activates intracellular signaling. One third of primary sensory neurons that bind to lectin IB4 do not express Trk receptors but the GDNF specific receptor Ret (Lin et al., 1993). GFRα-1 and RET are expressed in subpopulations of both small and large diameter DRG neurons, while GFRα-2 and GFRα-3 are found only in small diameter primary sensory neurons (Bennett et al., 2000). GFRα-1 is also expressed in glial cells (Trupp et al., 1997). Dramatic changes are found after axotomy in the expression of these receptors: GFRα-1 and Ret are greatly increased in large diameter neurons, whereas GFRα-3 increases in small diameter cells, and, in contrast, GFRα-2 is reduced after injury.

6.3. Fibroblast growth factor

Among the 23 members of the FGF family, FGF-2 has been shown to be the most important contributor to nerve regeneration (Grothe and Nikkhah, 2001). Different isoforms of this trophic factor are expressed in adult nervous tissue by glial cells and different neuronal populations, and play a key role in signal transduction in central and peripheral nervous systems (Ornitz and Itoh, 2001). Moreover, some FGF-2 isoforms seem to be differentially regulated during development and also after injury (Giordano et al., 1992; Grothe et al., 2000; Meisinger et al., 1996) and differential effects of FGF-2 isoforms have been reported. Transplantation of Schwann cells overexpressing the 21–23 kDa FGF-2 supported sensory recovery through a long gap injury, whereas the 18 kDa isoform inhibited myelination of regenerated axons (Haastert et al., 2006). In contrast, in cultured DRG neurons both low- and high molecular weight FGF-2 isoforms increase the number of axonal branches from control rats. Although it does not promote neurite elongation, FGF-2 significantly enhances the response to a preconditioning lesion (Klimaschewski et al., 2004). In vivo, FGF-2 is up-regulated both at the lesion site and in neuron bodies after nerve lesion. Whereas transgenic mice overexpressing FGF-2 showed greater number of regenerating axons after injury by regulating Schwann cell proliferation (Jungnickel et al., 2006), mutant mice lacking FGF-2 had significantly increased axon and myelin size but no difference in the number of regenerated fibers (Jungnickel et al., 2010). In fact, FGF-2 acts also on Schwann cells, as it was reported to stimulate their mitogenesis and proliferation (Davis and Stroobant, 1990).

Four different high affinity tyrosine transmembrane receptors have been described to which FGF-2 binds with different affinities (Ornitz et al., 1996). The main receptors in the nervous system, FGFR 1-3, are up-regulated after injury (Grothe et al., 1997) and in vitro studies found enhanced sensory neuron outgrowth after FGFR-1 overexpression, mediated by activation of extracellular signal-regulated kinase (ERK) and Akt pathways (Hausott et al., 2008). Moreover, FGF seems to maintain interactions with heparin and heparan sulfate protoglycan (HSPG) (Ornitz, 2000) that increase the affinity of the FGF-FGFR complex (McKeehan et al., 1998; Szebenyi and Fallon, 1999).

6.4. Insulin-like growth factors

IGF-1 and IGF-2 are related to the endogenous regulation of repair and regeneration processes. Both isoforms were reported to support motoneuron development as well as survival after axotomy when administered at the injury site (Pu et al., 1999). In vitro, addition of IGF-1 to the medium elicited neurite growth of sensory neurons, through activation of the PI3-kinase pathway (Kimpinski and Mearow, 2001). Moreover, exogenous application of this neurotrophic factor enhances the speed of axonal regeneration (Kanje et al., 1989; Glazner et al., 1993) and improves functional recovery (Emel et al., 2011). Some studies support the concept that IGF-1 promotes motor branch regeneration as it enhances motoneuron sprouting in vitro and in vivo (Caroni and Grandes, 1990), and it is up-regulated in ventral roots after injury (Hoke et al., 2006). Injured nerves show increased expression of IGF-1 and IGF-2 as well as IGF binding proteins (IGFBP) 4 and 5, whereas IGFBP-6 mRNA is strongly upregulated in spinal motoneurons after lesions, suggesting a special relevance for these neurons (Hammarberg et al., 1998). On the other hand, the expression of IGF-1 is known to decrease during aging and continuous application of this trophic factor improved axonal regeneration and muscle reinnervation in aging animals (Apel et al., 2010).

6.5. Neuregulins

Alternative splicing of the *NRG1* gene gives at least 16 different products (Falls, 2003; Steinthorsdottir et al., 2004) of which the most studied are neu differentiation factor (NDF), heregulin, acetylcholine receptor inducing activity (ARIA) and glial growth factor (GGF). NDF and heregulin are known to play a role in growth and differentiation activities in breast epithelial cells, and ARIA and GGF are found mainly at the neuromuscular junction and at Schwann cells during development (Corfas et al., 1993; Falls, 2003;

Holmes et al., 1992; Marchionni et al., 1993; Wen et al., 1992, 1994).

GGF is expressed by sensory, motor and sympathetic neurons (Marchionni et al., 1993; Chen and Ko, 1994). It has specific effects on Schwann cells, mediated via heterodimers of erbB2, erbB3 and erbB4 receptors, which seems to be important for the interactions between these glial cells and neurons. Following nerve lesion, GGF mRNA increases in DRG neurons, and expression of erbB2 and erbB3 receptors in glial cells of the distal nerve stump is also upregulated (Li et al., 1997) indicating a coordinated response to axotomy between axon and Schwann cells (Carroll et al., 1997). GGF promotes Schwann cell proliferation and maturation when regenerative axons have reinnervated target organs (Birchmeier and Nave, 2008; Falls, 2003; Raivich and Kreutzberg, 1994). GGF was shown to be a promising enhancer for glial cell proliferation when compared with NGF, as exogenous application of GGF after facial nerve anastomosis promotes Schwann cell migration and decreases the amount of myelin debris (Yildiz et al., 2011).

Neuregulin1 effect through erbB signaling is necessary for Schwann cell differentiation and myelination, establishment of neuromuscular junctions and also for normal sensory function development (Chen et al., 2006). Neuregulins may act as an axonally derived signal with trophic effect on denervated Schwann cells, facilitating their supportive role in axonal regeneration. In addition, neuregulins produced by Schwann cells themselves may be partially responsible for Schwann cell proliferation during Wallerian degeneration, via autocrine or paracrine mechanisms (Carroll et al., 1997). After sciatic nerve crush, neuregulin1 ablation resulted in a slower rate of regeneration, severe defects in remyelination, and abnormalities in neuromuscular junction reinnervation (Fricker et al., 2011).

6.6. Pleiotrophin

Pleiotrophin (PTN), also called heparin binding neurotrophic factor, is a heparin binding secreted protein. PTN is expressed in the nervous system during development and seems to trigger postsynaptic clustering of acetylcholine receptors (Peng et al., 1995). Two days after nerve injury there is an up-regulation of PTN, that reaches a peak after 7 days and returns back to baseline levels 3 months after injury (Mi et al., 2007). Immunohistochemical analysis showed PTN in nonneuronal cells in distal nerve segments, including Schwann cells, macrophages and endothelial cells, but not in axons (Blondet et al., 2005). High concentrations of this trophic factor were found in intact ventral roots, whereas they were low in intact cutaneous nerves. After injury it was dramatically upregulated in ventral roots, PTN being one of the neurotrophic factors more markedly overexpressed in axotomized motoneurons (Hoke et al., 2006). PTN application in spinal cord organotypic cultures promoted neurite elongation of motoneurons. Moreover, PTN increased axonal regeneration but not survival of avulsed spinal motoneurons (Chu et al., 2009). However, PTN applied to the lesioned peripheral nerve in vivo impaired muscle reinnervation (Blondet et al., 2006).

6.7. Osteopontin

Osteopontin (OPN) is a matricellular glycoprotein, strongly expressed by macrophages after injuries of the central but not in the peripheral nervous system (Küry et al., 2005), a fact that could explain why OPN is supposed to inhibit axonal growth only in the central nervous system. On the other hand, OPN is expressed by myelinating Schwann cells of intact peripheral nerves. One to 4 days after axotomy, OPN levels transiently increase in the distal nerve stump and there is a downregulation thereafter, reaching a minimum 14 days after the lesion (Jander et al., 2002). This event suggests a role of such glycoprotein in peripheral nerve regeneration, but this hypothesis needs further investigation to be corroborated.

6.8. Neuroactive cytokines

Some studies indicate that ciliary neurotrophic factor (CNTF) can be considered an "injury factor" released by glial cells after axotomy (Adler, 1993). It is known to be present in normal adult peripheral nerves, produced by the Schwann cells. Overexpression of CNTF in Schwann cells increases the expression of myelin proteins and activates their differentiation in vitro and in vivo (Homs et al., 2011). Released CNTF following nerve lesion may bind to the CNTF receptor a-ret, that is localized mainly in neurons and exerts a paracrine effect promoting cell survival and axonal growth (Hu et al., 2005). CNTF also promotes survival of motoneurons in neonatal animals following axotomy (Sendtner et al., 1990). This neuroactive cytokine seems to play an important role in guiding motoneuron reinnervation and in promoting the sprouting of their axons (Siegel et al., 2000). Indeed, systemic CNTF administration enhanced muscle fiber reinnervation and intramuscular nerve branching (Ulenkate et al., 1994). However, in the screening done by Hoke et al. (2006) CNTF was expressed at similar levels in intact cutaneous and motor nerves and its levels were similar in both sensory and motor axons after injury as well. In fact, viral-mediated overexpression of CNTF in Schwann cells enhanced the regenerative responses of both sensory and motor neurons (Homs et al., 2011).

The leukemia inhibitory factor (LIF) has an in vitro activation similar to CNTF in sympathetic neurons. It shares the signaling pathway with CNTF and it is considered that the roles of both cytokines are overlapping (Ip et al., 1992). Schwann cells are the main producer of LIF, that it is retrogradely transported by a subpopulation of small diameter neurons in DRG; most of them are positive for IB4, while the remaining are positive for TrkA and CGRP (Curtis et al., 1998; Thompson et al., 1997). Following nerve transection, LIF is up-regulated by Schwann cells at the injury site. In vivo studies suggest a role of LIF in promoting motor pathway regeneration, since its administration after nerve injury increases muscle mass and muscle contraction force (Tham et al., 1997). Moreover, motor regeneration is impaired in LIF knockout mice (Kurek et al., 1997). On the other hand, LIF seems to be essential for sympathetic (Rao et al., 1993) and sensory (Ekström et al., 2000; Cafferty et al., 2001) neuron survival and regeneration after injury.

Interleukin-6 (IL-6) and its receptor are upregulated by neurons after injury (Streit et al., 2000) and the combined over expression of both elements enhances nerve regeneration (Hirota et al., 1996). IL-6 also plays a role in the conditioning lesion effect, since in IL-6 $-/-$ mice there was a lack of GAP43 upregulation after a preconditioning injury and no enhancement of regeneration (Cafferty et al., 2004). However, the role of IL-6 in the conditioning lesion effect is controversial, since other authors defend that IL-6 is sufficient but not necessary to mimic the conditioning lesion effect on axonal growth (Cao et al., 2006). Moreover, the exogenous application of FGF-2 (low molecular weight isoform) strongly enhanced the mRNA levels of IL-6 and its receptor in Schwann cells, as both trophic factors are supposed to play a key role in the early reaction of Schwann cells to peripheral nerve injury (Grothe et al., 2000).

Although several experiments have demonstrated the usefulness of specific neurotrophic factors in nerve regeneration, it is becoming more apparent that the survival of sensory neurons and motoneurons depends on multiple neurotrophic factors acting synergistically or in a defined sequence (Terenghi, 1999). Furthermore, the possible preferential effects of some neurotrophic factors on regeneration of certain subpopulations of neurons have not been fully demonstrated in comprehensive comparative studies.

7. Extracellular matrix and adhesion molecules

Cell adhesion molecules (CAMs), extracellular matrix (ECM) molecules and guidance molecules maintain an important role during axonal regeneration, by exerting attraction/repulsion cues to the tip of the re-growing axon. Thanks to the receptors present at the growth cone site, the axons can respond to the different cues (Huber et al., 2003). For instance, chemoattraction is the consequence of a gradient that the growth cone interprets as a "signal" and converts into motion, giving rise to a well defined machinery in which ECM and CAM bindings are fundamental.

7.1. ECM components

The ECM is part of the extracellular environment in animal tissues, and it is composed of different kinds of glycoproteins, proteoglycans and also non-proteoglycan polysaccharides (as hyaluronic acid). It has mainly structural functions, providing support and maintaining cellular regulation. Many cells are able to bind the extracellular matrix and this cell-to-ECM adhesion is regulated by specific ECM receptors that are present on the cell surface.

In the peripheral nerve, Schwann cells and fibroblasts produce the components of the endoneurial ECM under the control of axons. The molecules present at the ECM may have stimulatory or inhibitory cues for axonal regeneration. Therefore, in an intact adult nerve, the "inhibitory" component predominates to prevent collateral sprouting, whereas after injury the ECM undergoes changes to favor a pro-regenerative environment. The ECM components can be divided into two main categories: proteoglycans and glycoproteins. The latter can be divided into collagen and non-collagenous molecules (for example tenascin, laminin and fibronectin) (Carbonetto et al., 1987).

Proteoglycans, such as chondroitin sulfate proteoglycans (CSPG), are found in the peripheral nerve, where they can inhibit the growth-promoting activities of other extracellular matrix components (Zuo et al., 1998). For example, they inhibit growing of the axons on laminin by interfering with integrin signaling (Hamel et al., 2008). Degradation of CSPG in the distal stump increases regeneration (Krekoski et al., 2001; Zuo et al., 2002) of both motor and sensory neurons (Udina et al., 2010). After injury, CSPGs are cleaved by activated matrix metalloproteinases (Ferguson and Muir, 2000; Zuo et al., 1998, 2002) and this allows axonal regeneration, creating a permissive environment. On the other hand, glycosaminoglycans (GAGs) have an enhancing role on in vitro neuritogenesis and in vivo nerve regeneration and muscle reinnervation. Moreover, GAGs are supposed to maintain interaction with IGF-1(Gorio et al., 1998).

Among the glycoproteins, three of them have an important role in axonal regeneration and their effects have been proved both in vivo and in vitro: collagen (Babington et al., 2005), fibronectin (Gardiner et al., 2007) and laminin (Werner et al., 2000).

Collagens are trimeric molecules formed by three α chains (for review see Gordon and Hahn, 2010) and can be divided into several sub-families according to the type of structure they form: fibrillar (types I, II, III, V, XI), facit-fibril associated collagens with interrupted triple helices (types IX, XII, XIV), short chain (types VIII, X), basement membrane (type IV) and other kinds with a different conformation (types VI, VII, XIII). Several studies showed the important role of collagens in axon pathfinding and in synaptic connection and maintenance (Fox, 2008).

Fibronectins are dimers formed by two almost identical monomers; in vertebrates, different kinds of isoforms are present due to alternative mRNA splicing. Fibronectin exists in two conformations: soluble and insoluble, the soluble fibronectin is one of the most abundant components of blood plasma and it is produced by hepatocytes, while the insoluble cellular fibronectin is incorporated in the double layer membrane of many cells types (Pankov and Yamada, 2002). In the nervous system fibronectin is secreted by Schwann cells and also by fibroblasts (Baron-Van Evercooren et al., 1986; Chernousov and Carey, 2000). It forms a fibrillar network and maintains interactions with collagen IV and laminins, promoting in this way cell proliferation and differentiation. Moreover, fibronectin binds mainly integrins but contains also domains for fibrin, collagen, heparin and syndecan (Mao and Schwarzbauer, 2005). Fibronectin is up-regulated immediately after peripheral nerve injury (Lefcort et al., 1992); embryonic isoforms of this ECM molecule were found to be up-regulated after injury and they seem to promote regeneration in different ways (Lefcort et al., 1996; Mathews and Ffrench-Constant, 1995; Vogelezang et al., 1999). However, in vitro studies demonstrated that fibronectin stimulates less neurite outgrowth in primary sensory neurons than laminin (Gardiner et al., 2007).

Laminins are heterotrimers consisting of α, β and γ chains. In vertebrates five α, three β and three γ chains have been found and these subunits can form 45 different laminin isoforms with a tissue-specific localization. The different isoforms are usually identified by changes in the α subunit: $\alpha1$ chain is expressed by developing and some adult epithelial cells; $\alpha2$ is mainly found in Schwann cells and basement membranes of striated muscles; $\alpha3$ is expressed in the epidermis; $\alpha4$ can be found in a variety of tissues, as endothelium, smooth muscles, fat cells, Schwann cells, neuromuscular junction and bone marrow, and $\alpha5$ is widely expressed in embryonic, developing and adult tissues (epithelia, endothelia, muscles, neuromuscular junctions) (Durbeej, 2010; Hallmann et al., 2005; Miner, 2008; Schéele et al., 2007; Tzu and Marinkovich, 2008). Some laminin isoforms are now known to play a fundamental role in nerve regeneration. In the peripheral nervous system, two different kinds of laminins are mainly present, containing $\alpha2$ and $\alpha4$ chains, both synthesized by Schwann cells (Wallquist et al., 2002), while $\alpha5$ can be detected in the sensory end organs (Caissie et al., 2006). Together with a $\beta1$ and a $\gamma1$ chain, these α subunits form laminin 2 ($\alpha2\beta1\gamma1$), laminin 8 ($\alpha4\beta1\gamma1$) and laminin 10 ($\alpha5\beta1\gamma1$), respectively. The main receptors for these isoforms are integrins $\alpha1\beta1$, $\alpha2\beta1$, $\alpha6\beta1$, $\alpha7\beta1$ and sydecans.

After nerve injury the levels of laminin 8 increase in the proximal stump, with a peak of up-regulation at 3 days and normalization around 42 days after the lesion, a fact that highlights the important role that this isoform probably plays in the early dynamics of nerve regeneration. On the other hand, laminin 2 is up-regulated after injury and this increase lasts longer than 42 days (Wallquist et al., 2002). The same study proposes a correlation between the expression of cytokines by macrophages and the up-regulation of laminins. During the early stages of nerve regeneration, there is a coordinated temporal expression of laminin $\alpha4$ and integrin $\alpha6$ subunits in relation to Schwann cell proliferation, while laminin $\alpha2$ and integrin $\alpha7$ subunits are over-expressed at later stages, playing a more direct function on axonal regeneration itself. Although separate studies point out that spinal motoneurons prefer to grow on substrates containing $\alpha2$ laminin (Wallquist et al., 2005) whereas sensory neurons extended neurites preferentially on $\alpha4$ laminin (Fried et al., 2005), no conclusive data supports the specificity of laminins for some populations of axons. Furthermore, there is not a differential pattern of expression of both isoforms $\alpha2$ and $\alpha4$ in motor and sensory roots (Plantman et al., 2008).

Tube repair is an alternative technique to the classical suture or autograft repair of transected peripheral nerves, but it also offers a unique method to investigate and manipulate the local environment of axonal regeneration. Regeneration in tubular guides is dependent on the formation of a connective cable that bridges the gap between the nerve stumps. Inside the tube with the attached nerve stumps at the ends, an initial fibrin clot is formed, and provides a guiding surface for the ingrowth of fibroblasts and Schwann cells migrating from both proximal and distal stumps (Liu, 1992; Williams et al., 1983). The intratubular cable is then enriched with ECM components, mainly collagen fibrils longitudinally oriented, fibronectin and laminin. Regenerating axons then grow from the proximal nerve, directed by the Schwann cells in the intratubular connective cable. In vivo studies that used components of the naturally formed ECM, such as fibrin, fibronectin, hyaluronate, collagen and laminin-containing gels to prefill the lumen of the tube guide, have reported enhancement of peripheral nerve regeneration with respect to empty guides (Williams, 1987; Bailey et al., 1993; Chamberlain et al., 1998; Madison et al., 1988; Labrador et al., 1998). On the contrary, application of anti-laminin antibody or anti-laminin/collagen receptor to grafts or tubes inhibits axonal regeneration, whereas anti-fibronectin antibody has no effect (Wang et al., 1992). In comparative studies, it was found that axonal growth for longer distances and higher levels of target reinnervation occurred with a laminin matrix with respect to collagen, fibrin and hyaluronate containing gels (Labrador et al., 1998; Navarro et al., 1996). On the other hand, by using an aligned collagen or laminin matrix, the rate and the direction of axonal elongation is improved due to contact guidance with the fibrils aligned along the tube axis, mimicking the natural orientation of the endoneurial tubules (Ceballos et al., 1999; Chamberlain et al., 1998; Verdu et al., 2002). However, most of the studies investigating the influence of ECM components in vivo have focused on enhancing axonal regeneration across long gaps, and little is known on the differential effects in vivo regarding selective interaction with sensory and motor axons.

7.2. CAM components

Different CAMs can be found in the nervous system and are well known for their key role in binding molecules of the ECM and the surface of other adjacent cells. Cadherins, members of the immunoglobulin superfamily (IgSF), and integrins mediate the interactions between the ECM and other CAMs present on other cells.

7.2.1. Cadherins

Cadherins (calcium-dependent adherent proteins) mediate cell to cell adhesion and consist in a superfamily with almost 50 different members (Angst et al., 2001). The different cadherin subfamilies present different structures; the prototype contains single-pass transmembrane proteins, five cadherin repeats in the ectodomain and a highly conserved cytoplasmic region (Juliano, 2002). Among all the different subtypes, N-cadherin is the most studied in the nervous system, where it plays an important role in axon outgrowth and fasciculation and in synaptic formation and consolidation (Ranscht, 2000). These adhesion molecules contain two different domains, the juxtamembrane domain (JMD) and the C-terminal catenin-binding domain (CBD) that can bind to catenins and form cadherin–catenin complexes.

7.2.2. Ig CAMs

Among the different members of the immunoglobulin (Ig) super family the most studied in the nervous system are the neural CAMs (NCAMs). These molecules are characterized by the presence of Ig-like domains, bind mainly in a homophilic manner and trigger different processes, such as axon pathfinding and target recognition. NCAM seems to mediate neurite outgrowth through at least two main different pathways: FGFR phosphorylation and Fyn association. Several studies indicate that the disruption of these interactions affect MAPK activation and the consequent neurite outgrowth (Jessen et al., 2001; Niethammer et al., 2002; Saffell et al., 1997). Moreover, the polysilated form of NCAM is known to play a crucial role in regeneration of motor nerve branches (Franz et al., 2005) and in promoting preferential reinnervation of muscles, as motor neurons in NCAM (−/−) mice reinnervate motor and cutaneous pathways with equal preference. Polysialic acid (PSA) is a carbohydrate attached to the glycoprotein backbone of the NCAM molecule, and it is implicated in nervous system development. Besides, PSA seems to be differently upregulated in motoneurons after injury; in fact, some pools of motoneurons show a high increase in PSA expression and this seems to trigger the preferential motor reinnervation of the quadriceps femoris muscle (Franz et al., 2008). Repair of the transected femoral nerve by a tube guide containing a PSA mimetic peptide enhanced Schwann cell proliferation, remyelination of regnerated axons and locomotion recovery. These effects were likely mediated by NCAM through its interaction with FGFR (Mehanna et al., 2009).

Other types of CAMs, as for example L1 and L2, are known to affect axonal regeneration. However, their role is controversial since L1 ablation in Schwann cells was shown to enhance functional recovery after peripheral nerve injury, suggesting an inhibitory role of this molecule in Schwann cell proliferation after nerve damage (Guseva et al., 2009). On the other hand, L2 epitope interacts with the adhesion molecule HNK-1, a marker of motor Schwann cells. It has been suggested that HNK-1 participates in motor axon regeneration (Mears et al., 2003; Vrbova et al., 2009) and influence preferential motor reinnervation (see above).

7.2.3. Integrins

Integrins are glycosylated heterodimers formed by α and β subunits, which are mainly implicated in focal adhesion and focal complex formations (cell-to-basal lamina interactions). Integrins are non-covalently associated type I transmembrane proteins which have a cytoplasmic (β) and an extracellular (α) domain (Van der Flier and Sonnenberg, 2001; Hynes, 2002). In vertebrates, 18 α and 8 β subunits have been described and these can form 24 different integrin complexes (Takada et al., 2007). Depending on their functions and interactions, integrins can be grouped into different sub-families (Table 1). Certain integrin subunits are ligand-specific, while some ECM molecules as laminin and

Table 1
Types of integrins and their main specifications. For complete review see (Barczyk et al., 2010).

β1 Family	ECM components and NT factor relations	Cell type expression
$\alpha1\beta1$	Collagens/Sema7A VEGF/TGFβ/PDGF/MCP-1/IL-8	Endothelial cells/ fibroblasts
$\alpha2\beta1$	Collagens/E-cadherin/Endorepellin VEGF	Endothelial cells/ fibroblasts
$\alpha3\beta1$	Laminins/CSPG4	Epithelial cells
$\alpha4\beta1$	Fibronectin/VCAM1	Endothelial cells
$\alpha5\beta1$	Fibronectin/Endostatin	Fibroblasts
$\alpha6\beta1$	Laminin	Epithelial cells and glia
$\alpha7\beta1$	Laminin	Muscle cells and neurons
$\alpha8\beta1$	Fibronectin/Vitronectin/Nephronectin	Kidney and ureter bud epithelium
$\alpha9\beta1$	Fibronectin/VCAM1/Cytotactin/ Osteopontin/Tenascin-C/VEGF	Endothelial cells
$\alpha10\beta1$	Collagens	Chondrocytes
$\alpha11\beta1$	Collagens PDGF	Fibroblasts
$\alpha V\beta1$	Fibronectin/Vitronectin	T-cells

fibronectin can bind to different integrins (Hynes, 2002). Integrins are also receptors for members of the Ig superfamily, thus mediating cell-to-cell adhesion. The integrin β1 family is composed of 12 different heterodimers; the β1 subunit provides the bound to actin cytoskeleton in each of this heterodimers, while the α subunit determines the ECM molecule-ligand specificity. Schwann cells express three different integrins: α1β1, α6β1 and α6β4, which are known to bind both laminin 2 and 8. Integrin α1β1 seems to be expressed only by immature Schwann cells, while α6β1 is expressed also during adulthood, although it is thought that its main function could be induction of myelination during development (Wallquist et al., 2002).

Integrins have both mechanical and chemical functions at the same time and are involved in outside-in and inside-out signaling. Integrins control Rho protein activation and translocation of Rac1 and Cdc42 to the plasma membrane (Del Pozo et al., 2002), especially in specific membrane domains called lipid rafts (see below). Many integrins can form complexes with receptor protein tyrosine kinases (RPTKs) (Giancotti and Ruoslahti, 1999; Yamada and Even-Ram, 2002, Table 1).

After axotomy, neurons overexpress α4β1, α5β1, α6β1 and α7β1 integrins (Ekström et al., 2003; Kloss et al., 1999; Vogelezang et al., 2001; Wallquist et al., 2004; Werner et al., 2000) that are important to promote axonal regeneration (Fig. 6). Thus, embryonic and neonatal neurons can grow on inhibitory environments due to the elevated expression of integrins among other intrinsic factors (as elevated cAMP, see above) when compared to mature neurons (Lemons and Condic, 2008).

Integrin subunit α7 is considered crucial for axonal regeneration in peripheral nerves. For example, deletion of the gene that encodes for α7 drastically reduces the speed of motor axon regeneration and delays target reinnervation (Werner et al., 2000). Similarly, the poor regenerative response observed in knock out mice of c-jun could be related to the lack of up-regulation of α7 integrin subunits in these animals after injury (Raivich et al., 2004). In agreement with these observations, over-expression of α7 subunit in injured neurons increases axon regeneration (Condic, 2001). The conditioning lesion effect also depends on integrin over-expression by sensory neurons. Increased neurite growth in DRG neurons primed by a previous axotomy is mediated by α7 integrin in matrigel or collagen substrates (Ekström et al., 2003). In contrast, over-expression of α5 integrin mediates the enhancement of neurite outgrowth of conditioned neurons on fibronectin (Gardiner et al., 2007).

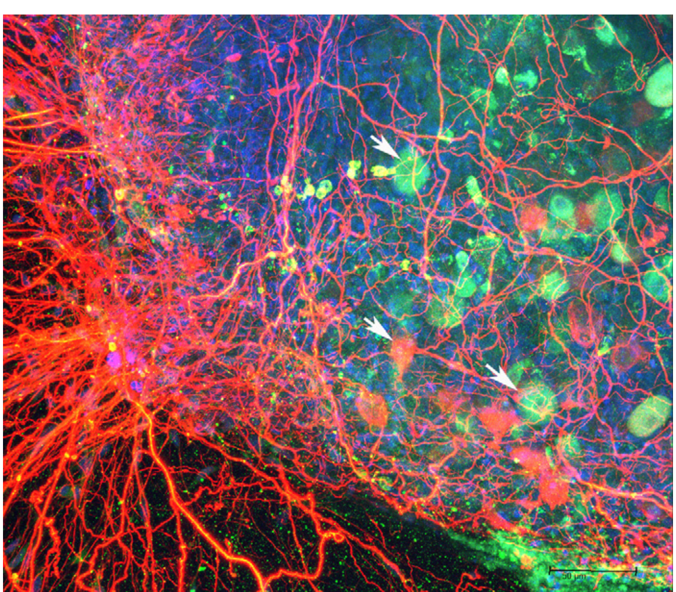

Fig. 6. In vitro expression of integrin β1 (green, indicated by the arrows) in regenerating sensory neurons of a DRG explant embedded in a collagen type I 3D matrix. The concentration of certain ligands influences the expression of integrin receptors and the elongation of neurites. Phosphorylated neurofilament – neuronal marker in red, DAPI – nuclear marker in blue.

The intracellular signals triggered by the interaction between ECM and integrins also influences the cellular response to soluble growth factors and cytokines. In fact, integrin-mediated cell adhesion not only initiates direct signals but also modulates transmission of signals downstream of growth factor receptors (Del Pozo et al., 2004, 2002; Schwartz and Ginsberg, 2002). Integrins, as other cell adhesion molecules, are necessary partners of growth factors and cytokine receptors (Giancotti and Tarone, 2003). For example, integrin β1 is involved in transmembrane signaling of GDNF, since the GDNF receptor GFRα-1 can form complexes with this integrin subunit after administration of GDNF (Cao et al., 2008), and GDNF can also increase the integrin levels (Chao et al., 2003; Funahashi et al., 2003; Grabham and Goldberg, 1997).

8. Interactions between neurotrophic factors, receptors and CAMs at lipid rafts

Interactions between cell adhesion molecules, growth factors and cytokines are mainly performed in specific plasma membrane domains called lipid rafts, an important structure rich in receptors that can be found at the growth cones.

The membrane bilayer in eukaryotic cells is mainly composed of three different classes of lipids: glycerophospholipids, sphingolipids and sterols; these kinds of lipids allow proteins to associate to the membrane thanks to their different properties. However, the lipid distribution in the cell membrane is asymmetrical and the composition and the physical state of certain lipid domains differ from the rest of the bilayer mosaic. Lipid rafts are rich in cholesterol and sphingolipids, and their state reveal a dynamic structure where cholesterol maintains a liquid order thanks to interaction with sphingolipids. A general marker of lipid rafts is the ganglioside GM1. Some lipid rafts are enriched with caveolin, and they are called caveolae microdomains. Among the proteins that accumulate in the lipid rafts are src-family kinases and G-proteins (Simons and Toomre, 2000). It is now known that lipid rafts are crucial for growth-factor signal transduction, cellular adhesion, axon guidance, vesicular trafficking, synaptic transmission and membrane-associated proteolysis (Tsui-Pierchala et al., 2002, Fig. 7).

Growth factors act mostly by binding to receptor tyrosine kinases, such as Trks, EGFR (epidermal growth factor receptor), PDGFR (platelet-derived growth factor receptor) and insulin receptor, which can be found in lipid rafts where they likely start intracellular signaling pathways (Couet et al., 1997; Huang et al., 1999; Liu and Dreyfuss, 1996; Mineo et al., 1996; Peiró et al., 2000; Wu et al., 1997; Yamamoto et al., 1998). Other receptors, like RET, translocate to the lipid rafts after or during activation (Paratcha et al., 2001; Tansey et al., 2000). GFRα-1 is also present in the lipid rafts (Poteryaev et al., 1999; Trupp et al., 1999) and it recruits RET after GDNF administration. Thus, any corruptive manipulations of the lipid raft blocks RET recruitment (Tansey et al., 2000).

TrkB is translocated at the lipid rafts after BDNF administration but its translocation has been related to short-term BDNF actions, as synaptic transmission and plasticity, whereas it does not affect neuronal survival (Suzuki et al., 2004). Similarly, Guirland et al. (2004) described a selective implication of lipid rafts in some of the functions of BDNF when examining its chemoattractant effect using a growth cone turning assay (Lohof et al., 1992). Disruption of lipid rafts by depletion or sequestration of cholesterol only affected the growth cone turning assay, while the growth cone rate

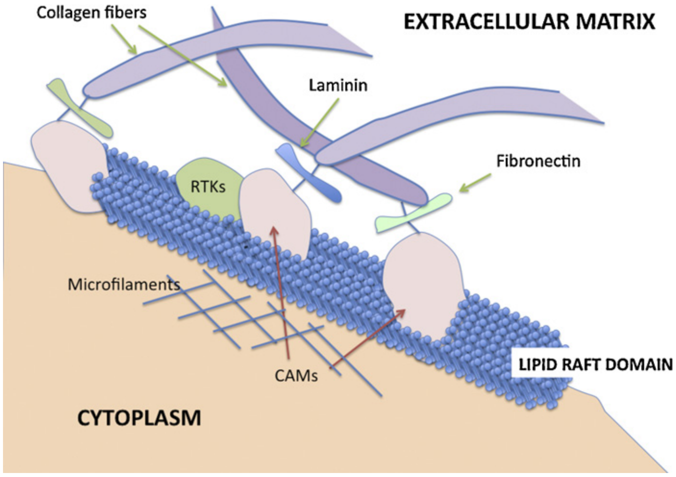

Fig. 7. The lipid raft is a microdomain platform rich in sphingolipids, cholesterol and specific membrane proteins such as GPI-anchored molecules. It is needed for fast signaling and plays an important role at the growth cone leading front, acting on cytoskeleton re-organization. Lipid rafts are enriched in growth factor receptor tyrosine kinases (RTKs) and CAMs, which can transduce their signals, interact and recruit other signaling molecules within the raft. The disruption of the lipid raft impairs the receptor activity, inhibiting downstream signals and axonal growth.

of extension mediated by BDNF was not altered. Thus, it seems that the lipid rafts can be mostly involved in specific behaviors of the cells. When the stability of lipid rafts was perturbed, by affecting GM1 gangliosides, BDNF effects on growth cone turning responses and elongation were abolished; therefore, it seems that gangliosides have an important role in BDNF signaling pathway, as reported also in other studies (Ferrari et al., 1995; Pitto et al., 1998). Repulsion from inhibiting molecules, such as netrin-1 and semaphorin 3A, was also blocked after the disruption of lipid rafts. Guirland et al. (2004) looked at TrkB receptor behavior and observed that, after BDNF administration just on one side of the growth cone, the receptors were distributed in an asymmetric way and were co-localized with gangliosides GM1 at the side of the growth cone that was exposed to the neurotrophic factor gradient. Therefore, the asymmetrical distribution of the lipid rafts in the cell may allow BDNF to elicit different effects under different conditions on the same neurons or just in certain parts of the cell (Guirland et al., 2004). On the other hand, the presence of P75 receptor in the lipid raft has an inhibitory role in the translocation of TrkB into the raft mediated by BDNF (Suzuki et al., 2004). The activated P75 receptor can also be localized in lipid rafts after NGF administration (Higuchi et al., 2003). Since in this study they used mouse cerebellar neurons and E18 rat hippocampal neurons that are known to express P75 but not TrkA, the results suggest that NGF induces translocation of phosphorylated P75 to lipid rafts in neurons that express only P75 as an NGF receptor.

Besides its role in growth factor signal transduction, lipid rafts play also a role in cell adhesion mechanisms: many adhesion molecules such as TAG-1 (transiently expressed axonal glycoprotein-1), NCAM-120 (neural cell adhesion molecule), Thy-1 and F3/contactin can be found in the lipid rafts and are GPI-anchored. Integrins have also been found in lipid rafts in fibroblasts, T cells and neurons (Brown and London, 1997; Ichikawa et al., 2009; Palazzo et al., 2004).

Thus, lipid rafts are key points where adhesion molecules and growth factors can interact and cross-talk. For example, the Rho family of small GTPase plays a central role in integrating signals from integrins and growth factors. Interestingly, the presence of Rho and Rac in lipid rafts depends on integrins. Integrins maintain lipid rafts in the plasma membrane, inhibit internalization of Rac binding sites and regulate Rho-mediated microtubule stabilization (Palazzo et al., 2004). Integrins can also control signal pathways other than Rho, like ras/Erk, JNK, PI-3-kinase, FAK and src family kinases, by raft internalization. ERK activation by integrins would be important in situations with low concentration of growth factors, where growth depends on co-stimulation of ERK by integrins and growth factors.

It is important to take into account the role of these structures in nerve regeneration when looking at the axonal rewiring, as they are known to make a specialized signaling platform in the plasma membrane, and therefore can transduce signals different from those in the non-raft membrane (Anderson and Jacobson, 2002; Simons and Toomre, 2000; Suzuki et al., 2004). Moreover, lipid rafts are a platform for protein complexes able to promote specific and short-lasting behaviors, for example, from neurotrophic factors and adhesion molecule signals, which will be affected by disruption of these microdomains.

9. Conclusions

Rewiring the peripheral nervous system after nerve injury has been considered an easier issue to manage when compared to the central nervous system repair. However, several mechanisms underlying nerve regeneration are not fully known yet. Several molecules are known to play an important role in axonal regeneration, both at the neuronal and at the distal nerve levels, but the way they interact between each other is still not clear. This issue reduces the possibilities of achieving optimal regeneration, also in terms of target organ specific reinnervation. The present review presents an overview of the factors that are playing a major role during peripheral nerve regeneration, taking in consideration the interactions between axons, glial cells and the extracellular matrix molecules. The switch of the gene machinery of the neurons to a pro-regenerative state triggers axon sprouting and elongation along the endoneurium, where Schwann cells have proliferated. Unfortunately, the topographic distribution of nerve fascicles and of motor and sensory fibers is disrupted after transection injuries of the nerve (Badia et al., 2010; Lago and Navarro, 2006), a fact that largely affects the accuracy of target organ reinnervation. Therefore, attempts to direct axons during re-growth and appropriate pathfinding will be crucial for improving functional recovery after severe nerve lesions. For such attempts, it is important to consider the interactions maintained between neurotrophic factor receptors and cell adhesion molecules, since their reciprocal influence is known to direct the tip of the axon during outgrowth. The emerging concept is to attempt re-wiring at the microdomain level, where neutrophic factor receptors and cell adhesion molecules are interacting and activating the regenerative machinery under the influence of the extracellular environment. Promising strategies to promote specific axon regeneration could be bio-engineered grafts to bridge the lesion gap (Rodriguez et al., 2000; Navarro et al., 2003; Marino et al., 2009), which combine the best substrates for regeneration, enriched with Schwann cells and funcionalized to over express selected trophic factors (Tannemaat et al., 2008; Mason et al., 2011).

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References

Abe, N., Borson, S.H., Gambello, M.J., Wang, F'., Cavalli, V., 2010. Mammalian target of rapamycin (mTOR) activation increases axonal growth capacity of injured peripheral nerves. Journal of Biological Chemistry 285 (36), 28034–28043.

Abe, N., Cavalli, V., 2008. Nerve injury signaling. Current Opinion in Neurobiology 18 (3), 276–283.

Adler, R., 1993. Ciliary neurotrophic factor as an injury factor. Current Opinion in Neurobiology 3 (5), 785–789.

Agthong, S., Kaewsema, A., Tanomsridejchai, N., Chentanez, V., 2006. Activation of MAPK ERK in peripheral nerve after injury. BMC Neuroscience 8, 45.

Airaksinen, M.S., Koltzenburg, M., Lewin, G.R., Masu, Y., Helbig, C., Wolf, E., Brem, G., Toyka, K.V., Thoenen, H., Meyer, M., 1996. Specific subtypes of cutaneous mechanoreceptors require neurotrophin-3 following peripheral target innervation. Neuron 16 (2), 287–295.

Anderson, R.G., Jacobson, K., 2002. A role for lipid shells in targeting proteins to caveolae, rafts, and other lipid domains. Science 296 (5574), 1821–1825.

Angst, B.D., Marcozzi, C., Magee, A.I., 2001. The cadherin superfamily: diversity in form and function. Journal of Cell Science 114 (Pt 4), 629–641.

Apel, P.J., Ma, J., Callahan, M., Northam, C.N., Alton, T.B., Sonntag, W.E., Li, Z., 2010. Effect of locally delivered IGF-1 on nerve regeneration during aging: an experimental study in rats. Muscle and Nerve 41 (3), 335–341.

Babington, E.J., Vatanparast, J., Verrall, J., Blackshaw, S.E., 2005. Three-dimensional culture of leech and snail ganglia for studies of neural repair. Invert Neuroscience 5 (3–4), 173–182.

Badia, J., Pascual-Font, A., Vivó, M., Udina, E., Navarro, X., 2010. Topographical distribution of motor fascicles in the sciatic-tibial nerve of the rat. Muscle and Nerve 42 (2), 192–201.

Bailey, S.B., Eichler, M.E., Villadiego, A., Rich, K.M., 1993. The influence of fibronectin and laminin during Schwann cell migration and peripheral nerve regeneration through silicone chambers. Journal of Neurocytology 22 (3), 176–184.

Baloh, R.H., Tansey, M.G., Lampe, P.A., Fahrner, T.J., Enomoto, H., Simburger, K.S., Leitner, M.L., Araki, T., Johnson Jr., E.M., Milbrandt, J., 1998. Artemin, a novel member of the GDNF ligand family, supports peripheral and central neurons and signals through the GFRalpha3-RET receptor complex. Neuron 21 (6), 1291–1302.

Barczyk, M., Carracedo, S., Gullberg, D., 2010. Integrins. Cell and Tissue Research 339 (1), 269–280.

Baron-Van Evercooren, A., Gansmüller, A., Gumpel, M., Baumann, N., Kleinman, H.K., 1986. Schwann cell differentiation in vitro: extracellular matrix deposition and interaction. Developmental Neuroscience 8 (3), 182–196.

Bennett, D.L., Averill, S., Clary, D.O., Priestley, J.V., McMahon, S.B., 1996a. Postnatal changes in the expression of the trkA high-affinity NGF receptor in primary sensory neurons. European Journal of Neuroscience 8 (10), 2204–2208.

Bennett, D.L., Boucher, T.J., Armanini, M.P., Poulsen, K.T., Michael, G.J., Priestley, J.V., Phillips, H.S., McMahon, S.B., Shelton, D.L., 2000. The glial cell line-derived neurotrophic factor family receptor components are differentially regulated within sensory neurons after nerve injury. Journal of Neuroscience 20 (1), 427–437.

Bennett, D.L., Dmietrieva, N., Priestley, J.V., Clary, D., McMahon, S.B., 1996b. trkA, CGRP and IB4 expression in retrogradely labelled cutaneous and visceral primary sensory neurones in the rat. Neuroscience Letters 206 (1), 33–36.

Benowitz, L.I., Lewis, E.R., 1983. Increased transport of 44,000- to 49,000-dalton acidic proteins during regeneration of the goldfish optic nerve: a two-dimensional gel analysis. Journal of Neuroscience 3 (11), 2153–2163.

Birchmeier, C., Nave, K.A., 2008. Neuregulin-1, a key axonal signal that drives Schwann cell growth and differentiation. Glia 56 (14), 1491–1497.

Blondet, B., Carpentier, G., Ferry, A., Courty, J., 2006. Exogenous pleiotrophin applied to lesioned nerve impairs muscle reinnervation. Neurochemical Research 31 (7), 907–913.

Blondet, B., Carpentier, G., Lafdil, F., Courty, J., 2005. Pleiotrophin cellular localization in nerve regeneration after peripheral nerve injury. Journal of Histochemistry and Cytochemistry 53 (8), 971–977.

Bodine-Fowler, S.C., Meyer, R.S., Moskovitz, A., Abrams, R., Botte, M.J., 1997. Inaccurate projection of rat soleus motoneurons: a comparison of nerve repair techniques. Muscle and Nerve 20 (1), 29–37.

Borke, R.C., Curtis, M., Ginsberg, C., 1993. Choline acetyltransferase and calcitonin gene-related peptide immunoreactivity in motoneurons after different types of nerve injury. Journal of Neurocytology 22 (3), 141–153.

Bouquet, C. and Nothias, F., 2007. Molecular mechanisms of axonal growth. In: Bagnard, D. (Ed): Axon growth and Guidance. Adv. Exp. Med. Biol. 621, 1–16.

Boyd, J.G., Gordon, T., 2001. The neurotrophin receptors, trkB and p75, differentially regulate motor axonal regeneration. Journal of Neurobiology 49 (4), 314–325.

Boyd, J.G., Gordon, T., 2003. Glial cell line-derived neurotrophic factor and brain-derived neurotrophic factor sustain the axonal regeneration of chronically axotomized motoneurons in vivo. Experimental Neurolgy 183 (2), 610–619.

Bradley, W.G., Asbury, A.K., 1970. Duration of synthesis phase in neurilemma cells in mouse sciatic nerve regeneration. Experimental Neurolgy 26 (2), 275–282.

Braun, S., Croizat, B., Lagrange, M.C., Warter, J.M., Poindron, P., 1996. Neurotrophins increase motoneurons' ability to innervate skeletal muscle fibers in rat spinal cord–human muscle cocultures. Journal of Neurological Sciences 136 (1-2), 17–23.

Braunewell, K.H., Martini, R., LeBaron, R., Kresse, H., Faissner, A., Schmitz, B., Schachner, M., 1995. Up-regulation of a chondroitin sulphate epitope during regeneration of mouse sciatic nerve: evidence that the immunoreactive molecules are related to the chondroitin sulphate proteoglycans decorin and versican. European Journal of Neuroscience 7 (4), 792–804.

Brown, D.A., London, E., 1997. Structure of detergent-resistant membrane domains: does phase separation occur in biological membranes? Biochemical and Biophysical Research Communications 240 (1), 1–7.

Brushart, T.M., 1988. Preferential reinnervation of motor nerves by regenerating motor axons. Journal of Neuroscience 8 (3), 1026–1031.

Brushart, T.M., 1993. Motor axons preferentially reinnervate motor pathways. Journal of Neuroscience 13 (6), 2730–2738.

Brushart, T.M., Gerber, J., Kessens, P., Chen, Y.G., Royall, R.M., 1998. Contributions of pathway and neuron to preferential motor reinnervation. Journal of Neuroscience 18 (21), 8674–8681.

Buj-Bello, A., Buchman, V.L., Horton, A., Rosenthal, A., Davies, A.M., 1995. GDNF is an age-specific survival factor for sensory and autonomic neurons. Neuron 15 (4), 821–828.

Bunge, M.B., Williams, A.K., Wood, P.M., 1982. Neuron-Schwann cell interaction in basal lamina formation. Developmental Biology 92 (2), 449–460.

Cafferty, W.B., Gardiner, N.J., Das, P., Qiu, J., McMahon, S.B., Thompson, S.W., 2004. Conditioning injury-induced spinal axon regeneration fails in interleukin-6 knock-out mice. Journal of Neuroscience 24 (18), 4432–4443.

Cafferty, W.B., Gardiner, N.J., Gavazzi, I., Powell, J., McMahon, S.B., Heath, J.K., Munson, J., Cohen, J., Thompson, S.W., 2001. Leukemia inhibitory factor determines the growth status of injured adult sensory neurons. Journal of Neuroscience 21 (18), 7161–7170.

Cai, D., Qiu, J., Cao, Z., McAtee, M., Bregman, B.S., Filbin, M.T., 2001. Neuronal cyclic AMP controls the developmental loss in ability of axons to regenerate. Journal of Neuroscience 21 (13), 4731–4739.

Caissie, R., Gingras, M., Champigny, M.F., Berthod, F., 2006. In vivo enhancement of sensory perception recovery in a tissue-engineered skin enriched with laminin. Biomaterials 27 (15), 2988–2993.

Cao, J.P., Yu, J.K., Li, C., Sun, Y., Yuan, H.H., Wang, H.J., Gao, D.S., 2008. Integrin beta1 is involved in the signaling of glial cell line-derived neurotrophic factor. Journal of Comparative Neurology 509 (2), 203–210.

Cao, Z., Gao, Y., Bryson, J.B., Hou, J., Chaudhry, N., Siddiq, M., Martinez, J., Spencer, T., Carmel, J., Hart, R.B., Filbin, M.T., 2006. The cytokine interleukin-6 is sufficient but not necessary to mimic the peripheral conditioning lesion effect on axonal growth. Journal of Neuroscience 26 (20), 5565–5573.

Carbonetto, S., Evans, D., Cochard, P., 1987. Nerve fiber growth in culture on tissue substrata from central and peripheral nervous systems. Journal of Neuroscience 7 (2), 610–620.

Caroni, P., Grandes, P., 1990. Nerve sprouting in innervated adult skeletal muscle induced by exposure to elevated levels of insulin-like growth factors. Journal of Cell Biology 110 (4), 1307–1317.

Carroll, S.L., Miller, M.L., Frohnert, P.W., Kim, S.S., Corbett, J.A., 1997. Expression of neuregulins and their putative receptors, ErbB2 and ErbB3, is induced during Wallerian degeneration. Journal of Neuroscience 17 (5), 1642–1659.

Ceballos, D., Navarro, X., Dubey, N., Wendelschafer-Crabb, G., Kennedy, W.R., Tranquillo, R.T., 1999. Magnetically aligned collagen gel filling a collagen nerve guide improves peripheral nerve regeneration. Experimental Neurolgy 158 (2), 290–300.

Condic, M., 2001. Adult neuronal regeneration induced by transgenic integrin expression. Journal of Neuroscience 21 (13), 4782–4788.

Corfas, G., Falls, D.L., Fischbach, G.D., 1993. ARIA, a protein that stimulates acetylcholine receptor synthesis, also induces tyrosine phosphorylation of a 185-kDa muscle transmembrane protein. Proceedings of the National Academy of Sciences of the United States of America 90 (4), 1624–1628.

Costigan, M., Befort, K., Karchewski, L., Griffin, R.S., D'Urso, D., Allchorne, A., Sitarski, J., Mannion, J.W., Pratt, R.E., Woolf, C.J., 2002. Replicate high-density rat genome oligonucleotide microarrays reveal hundreds of regulated genes in the dorsal root ganglion after peripheral nerve injury. BMC Neuroscience 3, 16.

Couet, J., Sargiacomo, M., Lisanti, M.P., 1997. Interaction of a receptor tyrosine kinase, EGF-R, with caveolins. Caveolin binding negatively regulates tyrosine and serine/threonine kinase activities. Journal of Biological Chemistry 272 (48), 30429–30438.

Curtis, R., Tonra, J.R., Stark, J.L., Adryan, K.M., Park, J.S., Cliffer, K.D., Lindsay, R.M., DiStefano, P.S., 1998. Neuronal injury increases retrograde axonal transport of the neurotrophins to spinal sensory neurons and motor neurons via multiple receptor mechanisms. Molecular and Cellular Neuroscience 12 (3), 105–118.

Chamberlain, L.J., Yannas, I.V., Hsu, H.P., Stritchartz, G., Spector, M., 1998. Collagen-GAG substrate enhances the quality of nerve regeneration through collagen tubes up to level of autograft. Experimental Neurolgy 154 (2), 315–329.

Chao, C.C., Ma, Y.L., Chu, K.Y., Lee, E.H., 2003. Integrin alphav and NCAM mediate the effects of GDNF on DA neuron survival, outgrowth, DA turnover and motor activity in rats. Neurobiology of Aging 24 (1), 105–116.

Chen, L., Ko, C.P., 1994. Extension of synaptic extracellular matrix during nerve terminal sprouting in living frog neuromuscular junctions. Journal of Neuroscience 14 (2), 796–808.

Chen, S., Velardez, M.O., Warot, X., Yu, Z.X., Miller, S.J., Cros, D., Corfas, G., 2006. Neuregulin 1-erbB signaling is necessary for normal myelination and sensory function. Journal of Neuroscience 26 (12), 3079–3086.

Chernousov, M.A., Carey, D.J., 2000. Schwann cell extracellular matrix molecules and their receptors. Histology and Histopathology 15 (2), 593–601.

Christie, K.J., Webber, C.A., Martinez, J.A., Singh, B., Zochodne, D.W., 2010. PTEN inhibition to facilitate intrinsic regenerative outgrowth of adult peripheral axons. Journal of Neuroscience 30 (27), 9306–9315.

Chu, T.H., Li, S.Y., Guo, A., Wong, W.M., Yuan, Q., Wu, W., 2009. Implantation of neurotrophic factor-treated sensory nerve graft enhances survival and axonal regeneration of motoneurons after spinal root avulsion. Journal of Neuropathology and Experimental Neurology 68 (1), 94–101.

Davis, J.B., Stroobant, P., 1990. Platelet-derived growth factors and fibroblast growth factors are mitogens for rat Schwann cells. Journal of Cell Biology 110 (4), 1353–1360.

Del Pozo, M.A., Alderson, N.B., Kiosses, W.B., Chiang, H.H., Anderson, R.G., Schwartz, M.A., 2004. Integrins regulate Rac targeting by internalization of membrane domains. Science 303 (5659), 839–842.

Del Pozo, M.A., Kiosses, W.B., Alderson, N.B., Meller, N., Hahn, K.M., Schwartz, M.A., 2002. Integrins regulate GTP-Rac localized effector interactions through dissociation of Rho-GDI. Nature Cell Biology 4 (3), 232–239.

Del Río, J.A., González-Billault, C., Ureña, J.M., Jiménez, E.M., Barallobre, M.J., Pascual, M., Pujadas, L., Simó, S., La Torre, A., Wandosell, F., Avila, J., Soriano, E., 2004. MAP1B is required for Netrin 1 signaling in neuronal migration and axonal guidance. Current Biology 14 (10), 840–850.

Diamond, J., Coughlin, M., Macintyre, L., Holmes, M., Visheau, B., 1987. Evidence that endogenous beta nerve growth factor is responsible for the collateral sprouting, but not the regeneration, of nociceptive axons in adult rats. Proceedings of the National Academy of Sciences of the United States of America 84 (18), 6596–6600.

Dickson, B., 2002. Molecular mechanisms of axon guidance. Science 298 (5600), 1959–1964.

Domeniconi, M., Filbin, M.T., 2005. Overcoming inhibitors in myelin to promote axonal regeneration. Journal of Neurological Sciences 233 (1–2), 43–47.

Dubovy, P., 2004. Schwann cells and endoneurial extracellular matrix molecules as potential cues for sorting of regenerated axons: a review. Anatomical Science International 79 (4), 198–208.

Dumoulin, F.L., Raivich, G., Streit, W.J., Kreutzberg, G.W., 1991. Differential regulation of calcitonin gene-related peptide (CGRP) in regenerating rat facial nucleus and dorsal root ganglion. European Journal of Neuroscience 3 (4), 338–342.

Durbeej, M., 2010. Laminins. Cell and Tissue Research 339 (1), 259–268.

Ebendal, T., Tomac, A., Hoffer, B.J., Olson, L., 1995. Glial cell line-derived neurotrophic factor stimulates fiber formation and survival in cultured neurons from peripheral autonomic ganglia. Journal of Neuroscience Research 40 (2), 276–284.

Ekström, P.A., Kerekes, N., Hökfelt, T., 2000. Leukemia inhibitory factor null mice: unhampered in vitro outgrowth of sensory axons but reduced stimulatory potential by nerve segments. Neuroscience Letters 281 (2-3), 107–110.

Ekström, P.A., Mayer, U., Panjwani, A., Pountney, D., Pizzey, J., Tonge, D.A., 2003. Involvement of alpha7beta1 integrin in the conditioning-lesion effect on sensory axon regeneration. Molecular and Cellular Neuroscience 22 (3), 383–395.

Emel, E., Ergün, S.S., Kotan, D., Gürsoy, E.B., Parman, Y., Zengin, A., Nurten, A., 2011. Effects of insulin-like growth factor-I and platelet-rich plasma on sciatic nerve crush injury in a rat model. Journal of Neurosurgery 114 (2), 522–528.

Eriksson, N.P., Aldskogius, H., Grant, G., Lindsay, R.M., Rivero-Melian, C., 1997. Effects of nerve growth factor, brain-derived neurotrophic factor and neurotrophin-3 on the laminar distribution of transganglionically transported choleragenoid in the spinal cord dorsal horn following transection of the sciatic nerve in the adult rat. Neuroscience 78 (3), 863–872.

Ernfors, P., Rosario, C.M., Merlio, J.P., Grant, G., Aldskogius, H., Persson, H., 1993. Expression of mRNAs for neurotrophin receptors in the dorsal root ganglion and spinal cord during development and following peripheral or central axotomy. Brain Research Molecular Brain Research 17 (3-4), 217–226.

Falls, D.L., 2003. Neuregulins: functions, forms, and signaling strategies. Experimental Cell Research 284 (1), 14–30.

Ferguson, T.A., Muir, D., 2000. MMP-2 and MMP-9 increase the neurite-promoting potential of schwann cell basal laminae and are upregulated in degenerated nerve. Molecular and Cellular Neuroscience 16 (2), 157–167.

Ferrari, G., Anderson, B.L., Stephens, R.M., Kaplan, D.R., Greene, L.A., 1995. Prevention of apoptotic neuronal death by GM1 ganglioside. Involvement of Trk neurotrophin receptors. Journal of Biological Chemistry 270 (7), 3074–3080.

Filbin, M.T., 2003. Myelin-associated inhibitors of axonal regeneration in the adult mammalian CNS. Nature Reviews Neuroscience 4 (9), 703–713.

Fine, E.G., Decosterd, I., Papaloïzos, M., Zurn, A.D., Aebischer, P., 2002. GDNF and NGF released by synthetic guidance channels support sciatic nerve regeneration across a long gap. European Journal of Neuroscience 15 (4), 589–601.

Fox, M.A., 2008. Novel roles for collagens in wiring the vertebrate nervous system. Current Opinion in Cell Biology 20 (5), 508–513.

Franz, C.K., Rutishauser, U., Rafuse, V.F., 2005. Polysialylated neural cell adhesion molecule is necessary for selective targeting of regenerating motor neurons. Journal of Neuroscience 25 (8), 2081–2091.

Franz, C.K., Rutishauser, U., Rafuse, V.F., 2008. Intrinsic neuronal properties control selective targeting of regenerating motoneurons. Brain 131 (Pt 6), 1492–1505.

Fricker, F.R., Lago, N., Balarajah, S., Tsantoulas, C., Tanna, S., Zhu, N., Fageiry, S.K., Jenkins, M., Garratt, A.N., Birchmeier, C., Bennett, D.L., 2011. Axonally derived neuregulin-1 is required for remyelination and regeneration after nerve injury in adulthood. Journal of Neuroscience 31 (9), 3225–3233.

Fried, K., Sime, W., Lillesaar, C., Virtanen, I., Tryggvasson, K., Patarroyo, M., 2005. Laminins 2 (alpha2beta1gamma1, Lm-211) and 8 (alpha4beta1gamma1, Lm-411) are synthesized and secreted by tooth pulp fibroblasts and differentially promote neurite outgrowth from trigeminal ganglion sensory neurons. Experimental Cell Research 307 (2), 329–341.

Fu, S.Y., Gordon, T., 1995a. Contributing factors to poor functional recovery after delayed nerve repair: prolonged axotomy. Journal of Neuroscience 15 (5 Pt 2), 3876–3885.

Fu, S.Y., Gordon, T., 1995b. Contributing factors to poor functional recovery after delayed nerve repair: prolonged denervation. Journal of Neuroscience 15 (5 Pt 2), 3886–3895.

Fujitani, M., Kawai, H., Proia, R.L., Kashiwagi, A., Yasuda, H., Yamashita, T., 2005. Binding of soluble myelin-associated glycoprotein to specific gangliosides induces the association of p75NTR to lipid rafts and signal transduction. Journal of Neurochemistry 94 (1), 15–21.

Fujitani, M., Yamagishi, S., Che, Y.H., Hata, K., Kubo, T., Ino, H., Tohyama, M., Yamashita, T., 2004. P311 accelerates nerve regeneration of the axotomized facial nerve. Journal of Neurochemistry 91 (3), 737–744.

Funahashi, H., Takeyama, H., Sawai, H., Furuta, A., Sato, M., Okada, Y., Hayakawa, T., Tanaka, M., Manabe, T., 2003. Alteration of integrin expression by glial cell line-derived neurotrophic factor (GDNF) in human pancreatic cancer cells. Pancreas 27 (2), 190–196.

Funakoshi, H., Frisén, J., Barbany, G., Timmusk, T., Zachrisson, O., Verge, V.M., Persson, H., 1993. Differential expression of mRNAs for neurotrophins and their receptors after axotomy of the sciatic nerve. Journal of Cell Biology 123 (2), 455–465.

Gardiner, N.J., Moffatt, S., Fernyhough, P., Humphries, M.J., Streuli, C.H., Tomlinson, D.R., 2007. Preconditioning injury-induced neurite outgrowth of adult rat sensory neurons on fibronectin is mediated by mobilisation of axonal alpha5 integrin. Molecular and Cellular Neuroscience 35 (2), 249–260.

Gavazzi, I., 2001. Semaphorin-neuropilin-1 interactions in plasticity and regeneration of adult neurons. Cell and Tissue Research 305 (2), 275–284.

Giancotti, F.G., Ruoslahti, E., 1999. Integrin signaling. Science 285 (5430), 1028–1032.

Giancotti, F.G., Tarone, G., 2003. Integrin signaling and cell fate. Annual Review of Cell and Devlopmental Biology 19, 173–206.

Gillen, C., Korfhage, C., Muller, H.W., 1997. Gene expression in nerve regeneration. Neuroscientist 3 (2), 112–122.

Giordano, S., Sherman, L., Lyman, W., Morrison, R., 1992. Multiple molecular weight forms of basic fibroblast growth factor are developmentally regulated in the central nervous system. Developmental Biology 152 (2), 293–303.

Glazner, G.W., Lupien, S., Miller, J.A., Ishii, D.N., 1993. Insulin-like growth factor II increases the rate of sciatic nerve regeneration in rats. Neuroscience 54 (3), 791–797.

Gloster, A., Diamond, J., 1992. Sympathetic nerves in adult rats regenerate normally and restore pilomotor function during an anti-NGF treatment that prevents their collateral sprouting. Journal of Comparative Neurology 326 (3), 363–374.

Gold, B., 1997. Axonal regeneration of sensory nerves is delayed by continuous intrathecal infusion of nerve growth factor. Neuroscience 76 (4), 1153–1158.

Gordon, M.K., Hahn, R.A., 2010. Collagens. Cell and Tissue Research 339 (1), 247–257.

Gordon, T., 2009. The role of neurotrophic factors in nerve regeneration. Neurosurgical Focus 26 (2), E3.

Gordon, T., 2010. The physiology of neural injury and regeneration: the role of neurotrophic factors. Journal of Communication Disorders 43 (4), 265–273.

Gordon, T., Sulaiman, O.A.R., Boyd, J.G., 2003. Experimental strategies to promote functional recovery after peripheral nerve injuries. Journal of the Peripheral Nervous System 8 (4), 236–250.

Gordon, T., Tyreman, N., Raji, M.A., 2011. The basis for diminished functional recovery after delayed peripheral nerve repair. Journal of Neuroscience 31 (14), 5325–5334.

Gordon-Weeks, P.R., Mansfield, S.G., 1992. Assembly of microtubules in growth cones. The role of microtubule-associated proteins. In: Letourneau, S.B., Macagno, E.R. (Eds.), The Nerve Growth Cone. Raven Press, New York, pp. 55–64.

Gorio, A., Vergani, L., Lesma, E., Di Giulio, A.M., 1998. Neuroprotection, neuroregeneration, and interaction with insulin-like growth factor-I: novel non-anticoagulant action of glycosaminoglycans. Journal of Neuroscience Research 51 (5), 559–562.

Grabham, P.W., Goldberg, D.J., 1997. Nerve growth factor stimulates the accumulation of beta1 integrin at the tips of filipodia in the growth cones of sympathetic neurons. Journal of Neuroscience 17 (14), 5455–5465.

Gramsbergen, A., IJkema-Paassen, J., Meek, M.F., 2000. Sciatic nerve transection in the adult rat: abnormal EMG patterns during locomotion by aberrant innervation of hindleg muscles. Experimental Neurolgy 161 (1), 183–193.

Grothe, C., Heese, K., Meisinger, C., Wewetzer, K., Kunz, D., Cattini, P., Otten, U., 2000. Expression of interleukin-6 and its receptor in the sciatic nerve and cultured Schwann cells: relation to 18-kD fibroblast growth factor-2. Brain Research 885 (2), 172–181.

Grothe, C., Meisinger, C., Hertenstein, A., Kurz, H., Wewetzer, K., 1997. Expression of fibroblast growth factor-2 and fibroblast growth factor receptor 1 messenger RNAs in spinal ganglia and sciatic nerve: regulation after peripheral nerve lesion. Neuroscience 76 (1), 123–135.

Grothe, C., Nikkhah, G., 2001. The role of basic fibroblast growth factor in peripheral nerve regeneration. Anatomy and Embryology (Berl.) 204 (3), 171–177.

Grothe, C., Unsicker, K., 1992. Basic fibroblast growth factor in the hypoglossal system: specific retrograde transport, trophic, and lesion-related responses. Journal of Neuroscience Research 32 (3), 317–328.

Groves, M.J., An, S.F., Giornetto, B., Scaravilli, F., 1999. Inhibition of sensory neuron apoptosis and prevention of loss by NT-3 administration following axotomy. Experimental Neurolgy 155 (2), 284–294.

Gu, Z.Z., Pan, Y.C., Cui, J.K., Klebuc, M.J., Shenaq, S., Liu, P.K., 1997. Gene expression and apoptosis in the spinal cord neurons after sciatic nerve injury. Neurochemistry International 30 (4-5), 417–426.

Guan, W., Puthenveedu, M.A., Condic, M.L., 2003. Sensory neuron subtypes have unique substratum preference and receptor expression before target innervation. Journal of Neuroscience 23 (5), 1781–1791.

Guirland, C., Suzuki, S., Kojima, M., Lu, B., Zheng, J.Q., 2004. Lipid rafts mediate chemotropic guidance of nerve growth cones. Neuron 42 (1), 51–62.
Guseva, D., Angelov, D.N., Irintchev, A., Schachner, M., 2009. Ablation of adhesion molecule L1 in mice favours Schwann cell proliferation and functional recovery after peripheral nerve injury. Brain 132 (Pt 8), 2180–2195.
Haastert, K., Lipokatic, E., Fischer, M., Timmer, M., Grothe, C., 2006. Differentially promoted peripheral nerve regeneration by grafted Schwann cells over-expressing different FGF-2 isoforms. Neurobiology of Disease 21 (1), 138–153.
Hall, S.M., 1986. Regeneration in cellular and acellular autografts in the peripheral nervous system. Neuropathology and Applied Neurobiology 12 (1), 27–46.
Hallmann, R., Horn, N., Selg, M., Wendler, O., Pausch, F., Sorokin, L.M., 2005. Expression and function of laminins in the embryonic and mature vasculature. Physiological Reviews 85 (3), 979–1000.
Hamel, M.G., Ajmo, J.M., Leonardo, C.C., Zuo, F., Sandy, J.D., Gottschall, P.E., 2008. Multimodal signaling by the ADAMTSs (a disintegrin and metalloproteinase with thrombospondin motifs) promotes neurite extension. Experimental Neurolgy 210 (2), 428–440.
Hammarberg, H., Risling, M., Hökfelt, T., Cullheim, S., Piehl, F., 1998. Expression of insulin-like growth factors and corresponding binding proteins (IGFBP 1-6) in rat spinal cord and peripheral nerve after axonal injuries. Journal of Comparative Neurology 400 (1), 57–72.
Hanz, S., Fainzilber, M., 2006. Retrograde signaling in injured nerve – the axon reaction revisited. Journal of Neurochemistry 99 (1), 13–19.
Hanz, S., Perlson, E., Willis, D., Zheng, J.Q., Massarwa, R., Huerta, J.J., Koltzenburg, M., Kohler, M., van-Minnen, J., Twiss, J.L., Fainzilber, M., 2003. Axoplasmic importins enable retrograde injury signaling in lesioned nerve. Neuron 40 (6), 1095–1104.
Hausott, B., Schlick, B., Vallant, N., Dorn, R., Klimaschewski, L., 2008. Promotion of neurite outgrowth by fibroblast growth factor receptor 1 overexpression and lysosomal inhibition of receptor degradation in pheochromocytoma cells and adult sensory neurons. Neuroscience 153 (2), 461–473.
Hay, N., Sonenberg, N., 2004. Upstream and downstream of mTOR. Genes and Development 18 (16), 1926–1945.
Henderson, C.E., Phillips, H.S., Pollock, R.A., Davies, A.M., Lemeulle, C., Armanini, M., Simmons, L., Moffet, B., Vandlen, R.A., Simpson, L.C., 1994a. GDNF: a potent survival factor for motoneurons present in peripheral nerve and muscle. Science 266 (5187), 1062–1064.
Herdegen, T., Kummer, W., Fiallos, C.E., Leah, J., Bravo, R., 1991. Expression of c-jun, jun b and jun d proteins in rat nervous system following transection of vagus nerve and cervical sympathetic trunk. Neuroscience 45 (2), 413–422.
Higuchi, H., Yamashita, T., Yoshikawa, H., Tohyama, M., 2003. PKA phosphorylates the p75 receptor and regulates its localization to lipid rafts. EMBO Journal 22 (8), 1790–1800.
Hirota, H., Kiyama, H., Kishimoto, T., Taga, T., 1996. Accelerated nerve regeneration in mice by upregulated expression of interleukin (IL) 6 and IL-6 receptor after trauma. Journal of Experimental Medicine 183 (6), 2627–2634.
Hoffman, P.N., Cleveland, D.W., 1988. Neurofilament and tubulin expression recapitulates the developmental program during axonal regeneration: induction of a specific beta-tubulin isotype. Proceedings of the National Academy of Sciences of the United States of America 85 (12), 4530–4533.
Hoffman, P.N., Cleveland, D.W., Griffin, J.W., Landes, P.W., Cowan, N.J., Price, D.L., 1987. Neurofilament gene expression: a major determinant of axonal caliber. Proceedings of the National Academy of Sciences of the United States of America 84 (10), 3472–3476.
Hoffman, P.N., Lasek, R.J., 1980. Axonal transport of the cytoskeleton in regenerating motor neurons: constancy and change. Brain Research 202 (2), 317–333.
Höke, A., Cheng, C., Zochodne, D.W., 2000. Expression of glial cell line-derived neurotrophic factor family of growth factors in peripheral nerve injury in rats. Neuroreport 11 (8), 1651–1654.
Hoke, A., Gordon, T., Zochodne, D.W., Sulaiman, O.A.R., 2002. A decline in glial cell-line-derived neurotrophic factor expression is associated with impaired regeneration after long-term Schwann cell denervation. Experimental Neurolgy 173 (1), 77–85.
Höke, A., Ho, T., Crawford, T.O., LeBel, C., Hilt, D., Griffin, J.W., 2003. Glial cell line-derived neurotrophic factor alters axon Schwann cell units and promotes myelination in unmyelinated nerve fibers. Journal of Neuroscience 23 (2), 561–567.
Hoke, A., Redett, R., Hameed, H., Jari, R., Zhou, C., Li, Z.B., Griffin, J.W., Brushart, T.M., 2006. Schwann cells express motor and sensory phenotypes that regulate axon regeneration. Journal of Neuroscience 26 (38), 9646–9655.
Hoke, A., Sun, H.S., Gordon, T., Zochodne, D.W., 2001. Do denervated peripheral nerve trunks become ischemic? The impact of chronic denervation on vasa nervorum. Experimental Neurolgy 172 (2), 398–406.
Hökfelt, T., Broberger, C., Xu, Z.Q., Sergeyev, V., Ubink, R., Diez, M., 2000. Neuropeptides – an overview. Neuropharmacology 39 (8), 1337–1356.
Holmes, W.E., Sliwkowski, M.X., Akita, R.W., Henzel, W.J., Lee, J., Park, J.W., Yansura, D., Abadi, N., Raab, H., Lewis, G.D., et al., 1992. Identification of heregulin, a specific activator of p185erbB2. Science 256 (5060), 1205–1210.
Homs, J., Ariza, L., Pages, G., Udina, E., Navarro, X., Chillón, M., Bosch, A., 2011. Schwann cell targeting via intrasciatic injection of AAV8 as gene therapy strategy for peripheral nerve regeneration. Gene Therapy 18 (6), 622–630.
Honig, M.G., Frase, P.A., Camilli, S.J., 1998. The spatial relationship among cutaneous, muscle sensory and motoneuron axons during development of the chick hindlimb. Development 125 (6), 995–1004.
Horch, K.W., Lisney, S.J.W., 1981. On the number and nature of regenerating myelinated axons after lesions of cutaneous nerves in the cat. Journal of Physiology 313, 275–286.
Hory-Lee, F., Russell, M., Lindsay, R.M., Frank, E., 1993. Neurotrophin 3 supports the survival of developing muscle sensory neurons in culture. Proceedings of the National Academy of Sciences of the United States of America 90 (7), 2613–2617.
Hu, Y., Leaver, S.G., Plant, G.W., Hendriks, W.T., Niclou, S.P., Verhaagen, J., 2005. Lentiviral mediated transfer of CNTF to Schwann cells within reconstructed peripheral nerve grafts enhances adult retinal ganglion cell survival and axonal regeneration. Molecular Therapy 11 (6), 906–915.
Huang, C.S., Zhou, J., Feng, A.K., Lynch, C.C., Klumperman, J., DeArmond, S.J., Mobley, W.C., 1999. Nerve growth factor signaling in caveolae-like domains at the plasma membrane. Journal of Biological Chemistry 274 (51), 36707–36714.
Huber, A.B., Kolodkin, A.L., Ginty, D.D., Cloutier, J.F., 2003. Signaling at the growth cone: ligand-receptor complexes and the control of axon growth and guidance. Annual Review of Neuroscience 26, 509–563.
Huber, K., Meisinger, C., Grothe, C., 1997. Expression of fibroblast growth factor-2 in hypoglossal motoneurons is stimulated by peripheral nerve injury. Journal of Comparative Neurology 382 (2), 189–198.
Hynes, R., 2002. Integrins: bidirectional, allosteric signaling machines. Cell 110 (6), 673–687.
Ichikawa, N., Iwabuchi, K., Kurihara, H., Ishii, K., Kobayashi, T., Sasaki, T., Hattori, N., Mizuno, Y., Hozumi, K., Yamada, Y., Arikawa-Hirasawa, E., 2009. Binding of laminin-1 to monosialoganglioside GM1 in lipid rafts is crucial for neurite outgrowth. Journal of Cell Science 122 (Pt 2), 289–299.
Ide, C., 1996. Peripheral nerve regeneration. Neuroscience Research 25 (2), 101–121.
Ip, N.Y., Nye, S.H., Boulton, T.G., Davis, S., Taga, T., Li, Y., Birren, S.J., Yasukawa, K., Kishimoto, T., Anderson, D.J., 1992. CNTF and LIF act on neuronal cells via shared signaling pathways that involve the IL-6 signal transducing receptor component gp130. Cell 69 (7), 1121–1132.
Jander, S., Bussini, S., Neuen-Jacob, E., Bosse, F., Menge, T., Müller, H.W., Stoll, G., 2002. Osteopontin: a novel axon-regulated Schwann cell gene. Journal of Neuroscience Research 67 (2), 156–166.
Jankowski, M.P., McIlwrath, S.L., Jing, X., Cornuet, P.K., Salerno, K.M., Koerber, H.R., Albers, K.M., 2009. Sox11 transcription factor modulates peripheral nerve regeneration in adult mice. Brain Research 1256, 43–54.
Jenq, C.-B., Coggeshall, R.E., 1985. Numbers of regenerating axons in parent and tributary peripheral nerves in the rat. Brain Research 326 (1), 27–40.
Jessen, K.R., Mirsky, R., 2008. Negative regulation of myelination: relevance for development, injury, and demyelinating disease. Glia 56 (14), 1552–1565.
Jessen, U., Novitskaya, V., Pedersen, N., Serup, P., Berezin, V., Bock, E., 2001. The transcription factors CREB and c-Fos play key roles in NCAM-mediated neuritogenesis in PC12-E2 cells. Journal of Neurochemistry 79 (6), 1149–1160.
Juliano, R.L., 2002. Signal transduction by cell adhesion receptors and the cytoskeleton: functions of integrins, cadherins, selectins, and immunoglobulin-superfamily members. Annual Review of Pharmacology and Toxicology 42, 283–323.
Jungnickel, J., Haase, K., Konitzer, J., Timmer, M., Grothe, C., 2006. Faster nerve regeneration after sciatic nerve injury in mice over-expressing basic fibroblast growth factor. Journal of Neurobiology 66 (9), 940–948.
Jungnickel, J., Haastert, K., Grzybek, M., Thau, N., Lipokatic-Takacs, E., Ratzka, A., Nölle, A., Claus, P., Grothe, C., 2010. Mice lacking basic fibroblast growth factor showed faster sensory recovery. Experimental Neurolgy 223 (1), 166–172.
Kanje, M., Skottner, A., Sjöberg, J., Lundborg, G., 1989. Insulin-like growth factor I (IGF-I) stimulates regeneration of the rat sciatic nerve. Brain Research 486 (2), 396–398.
Kashiba, H., Senba, E., 1999. Up- and down-regulation of BDNF mRNA in distinct subgroups of rat sensory neurons after axotomy. Neuroreport 10 (17), 3561–3565.
Kimpinski, K., Mearow, K., 2001. Neurite growth promotion by nerve growth factor and insulin-like growth factor-1 in cultured adult sensory neurons: role of phosphoinositide 3-kinase and mitogen activated protein kinase. Journal of Neuroscience Research 63 (6), 486–499.
Kirsch, M., Terheggen, U., Hofmann, H.D., 2003. Ciliary neurotrophic factor is an early lesion-induced retrograde signal for axotomized facial motoneurons. Molecular and Cellular Neuroscience 24 (1), 130–138.
Klein, R., Lamballe, F., Bryant, S., Barbacid, M., 1992. The trkB tyrosine protein kinase is a receptor for neurotrophin-4. Neuron 8 (5), 947–956.
Klein, R., Parada, L.F., Coulier, F., Barbacid, M., 1989. trkB, a novel tyrosine protein kinase receptor expressed during mouse neural development. EMBO Journal 8 (12), 3701–3709.
Klimaschewski, L., Nindl, W., Feurle, J., Kavakebi, P., Kostron, H., 2004. Basic fibroblast growth factor isoforms promote axonal elongation and branching of adult sensory neurons in vitro. Neuroscience 126 (2), 347–353.
Kloss, C.U., Werner, A., Klein, M.A., Shen, J., Menuz, K., Probst, J.C., Kreutzberg, G.W., Raivich, G., 1999. Integrin family of cell adhesion molecules in the injured brain: regulation and cellular localization in the normal and regenerating mouse facial motor nucleus. Journal of Comparative Neurology 411 (1), 162–178.
Kobayashi, N.R., Bedard, A.M., Hincke, M.T., Tetzlaff, W., 1996. Increased expression of BDNF and trkB mRNA in rat facial motoneurons after axotomy. European Journal of Neuroscience 8 (5), 1018–1029.
Krekoski, C.A., Neubauer, D., Zou, J., Muir, D., 2001. Axonal regeneration into acellular nerve grafts is enhanced by degradation of chondroitin sulfate proteoglycan. Journal of Neuroscience 21 (16), 6206–6213.

Krull, C.E., Koblar, S.A., 2000. Motor axon pathfinding in the peripheral nervous system. Brain Research Bulletin 53 (5), 479–487.
Kurek, J.B., Bower, J.J., Romanella, M., Koentgen, F., Murphy, M., Austin, L., 1997. The role of leukemia inhibitory factor in skeletal muscle regeneration. Muscle Nerve 20 (7), 815–822.
Küry, P., Zickler, P., Stoll, G., Hartung, H.P., Jander, S., 2005. Osteopontin, a macrophage-derived matricellular glycoprotein, inhibits axon outgrowth. FASEB Journal 19 (3), 398–400.
Labrador, R.O., Buti, M., Navarro, X., 1998. Influence of collagen and laminin gels concentration on nerve regeneration after resection and tube repair. Experimental Neurolgy 149 (1), 243–252.
Lago, N., Navarro, X., 2006. Correlation between target reinnervation and distribution of motor axons in the injured rat sciatic nerve. Journal of Neurotrauma 23 (2), 227–240.
Lago, N., Rodríguez, F.J., Guzmán, M.S., Jaramillo, J., Navarro, X., 2007. Effects of motor and sensory nerve transplants on amount and specificity of sciatic nerve regeneration. Journal of Neuroscience Research 85 (12), 2800–2812.
Lamballe, F., Klein, R., Barbacid, M., 1991. trkC, a new member of the trk family of tyrosine protein kinases, is a receptor for neurotrophin-3. Cell 66 (5), 967–979.
Landmesser, L.T., 2001. The acquisition of motoneuron subtype identity and motor circuit formation. International Journal of Developmental Neuroscience 19 (2), 175–182.
Leah, J.D., Herdegen, T., Bravo, R., 1991. Selective expression of Jun proteins following axotomy and axonal transport block in peripheral nerves in the rat: evidence for a role in the regeneration process? Brain Research 566 (561–562), 198–207.
Leclere, P.G., Norman, E., Groutsi, F., Coffin, R., Mayer, U., Pizzey, J., Tonge, D., 2007. Impaired axonal regeneration by isolectin B4-binding dorsal root ganglion neurons in vitro. Journal of Neuroscience 27 (5), 1190–1199.
Lee, A.F., Ho, D.K., Zanassi, P., Walsh, G.S., Kaplan, D.R., Miller, F.D., 2004. Evidence that DeltaNp73 promotes neuronal survival by p53-dependent and p53-independent mechanisms. Journal of Neuroscience 24 (41), 9174–9184.
Lefcort, F., Clary, D.O., Rusoff, A.C., Reichardt, L.F., 1996. Inhibition of the NT-3 receptor TrkC, early in chick embryogenesis, results in severe reductions in multiple neuronal subpopulations in the dorsal root ganglia. Journal of Neuroscience 16 (11), 3704–3713.
Lefcort, F., Venstrom, K., McDonald, J.A., Reichardt, L.F., 1992. Regulation of expression of fibronectin and its receptor, alpha 5 beta 1, during development and regeneration of peripheral nerve. Development 116 (3), 767–782.
Lemons, M.L., Condic, M.L., 2008. Integrin signaling is integral to regeneration. Experimental Neurolgy 209 (2), 343–352.
Letourneau, P.C., Shattuck, T.A., 1989. Distribution and possible interactions of actin-associated proteins and cell adhesion molecules of nerve growth cones. Development 105 (3), 505–519.
Levi-Montalcini, R., 1987. The nerve growth factor: thirty-five years later. Bioscience Reports 7 (9), 681–699.
Lewin, G.R., Barde, Y.A., 1996. Physiology of the neurotrophins. Annual Review of Neuroscience 19, 289–317.
Lewis, A.K., Bridgman, P.C., 1992. Nerve growth cone lamellipodia contain two populations of actin filaments that differ in organization and polarity. Journal of Cell Biology 119 (5), 1219–1243.
Li, H., Terenghi, G., Hall, S.M., 1997. Effects of delayed reinnervation on the expression of c-erbB receptors by chronically denervated rat Schwann cells in vivo. Glia 20 (4), 333–347.
Li, X.Q., Verge, V.M., Johnston, J.M., Zochodne, D.W., 2004. CGRP peptide and regenerating sensory axons. Journal of Neuropathology and Experimental Neurology 63 (10), 1092–1103.
Lin, L.F., Doherty, D.H., Lile, J.D., Bektesh, S., Collins, F., 1993. GDNF: a glial cell line-derived neurotrophic factor for midbrain dopaminergic neurons. Science 260 (5111), 1130–1132.
Lindwall, C., Kanje, M., 2005. Retrograde axonal transport of JNK signaling molecules influence injury induced nuclear changes in p-c-Jun and ATF3 in adult rat sensory neurons. Molecular and Cellular Neuroscience 29 (22), 269–282.
Liu, H.M., 1992. The role of extracellular matrix in peripheral nerve regeneration: a wound chamber study. Acta Neuropathologica 83 (5), 469–474.
Liu, K., Lu, Y., Lee, J.K., Samara, R., Willenberg, R., Sears-Kraxberger, I., Tedeschi, A., Park, K.K., Jin, D., Cai, B., Xu, B., Connolly, L., Steward, O., Zheng, B., He, Z., 2010. PTEN deletion enhances the regenerative ability of adult corticospinal neurons. Nature Neuroscience 13 (9), 1075–1081.
Liu, Q., Dreyfuss, G., 1996. A novel nuclear structure containing the survival of motor neurons protein. EMBO Journal 15 (14), 3555–3565.
Liu, R.Y., Snider, W.D., 2001. Different signaling pathways mediate regenerative versus developmental sensory axon growth. Journal of Neuroscience 21 (17), RC164.
Lohof, A.M., Quillan, M., Dan, Y., Poo, M.M., 1992. Asymmetric modulation of cytosolic cAMP activity induces growth cone turning. Journal of Neuroscience 12 (4), 1253–1261.
Lorber, B., Howe, M.L., Benowitz, L.I., Irwin, N., 2009. Mst3b, an Ste20-like kinase, regulates axon regeneration in mature CNS and PNS pathways. Nature Neuroscience 12 (11), 1407–1414.
Lotfi, P., Garde, K., Chouhan, A.K., Bengali, E., Romero-Ortega, M.I., 2011. Modality-specific axonal regeneration: toward selective regenerative neural interfaces. Frontiers in Neuroengineering 4, 11.
Lowery, L.A., Van Vactor, D., 2009. The trip of the tip: understanding the growth cone machinery. Nature Reviews Molecular Cell Biology 10 (5), 332–343.
Lundborg, G., 2003. Richard P. Bunge memorial lecture. Nerve injury and repair – a challenge to the plastic brain. Journal of the Peripheral Nervous System 8 (4), 209–226.
Lundborg, G., Rosen, B., Dahlin, L., Holmberg, J., Rosen, I., 2004. Tubular repair of the median or ulnar nerve in the human forearm: a 5-year follow-up. Journal of Hand Surgery Br. 29 (2), 100–107.
Luo, L., 2002. Actin cytoskeleton regulation in neuronal morphogenesis and structural plasticity. Annual Review of Cell and Devlopmental Biology 18, 601–635.
Ma, C.H., Omura, T., Cobos, E.J., Latremoliere, A., Ghasemlou, N., Brenner, G.J., van Veen, E., Barrett, L., Sawada, T., Gao, F., Coppola, G., Gertler, F., Costigan, M., Geschwind, D., Woolf, C.J., 2011. Accelerating axonal growth promotes motor recovery after peripheral nerve injury in mice. Journal of Clinical Investment 121 (11), 4332–4347.
Mackinnon, S.E., Dellon, A.L., O'Brien, J.P., 1991. Changes in nerve fiber numbers distal to a nerve repair in the rat sciatic nerve model. Muscle Nerve 14 (11), 1116–1122.
Madiai, F., Hussain, S.R., Goettl, V.M., Burry, R.W., Stephens, R.L.J., Hackshaw, K.V., 2003. Upregulation of FGF-2 in reactive spinal cord astrocytes following unilateral lumbar spinal nerve ligation. Experimental Brain Research 148 (3), 366–376.
Madison, R.D., Archibald, S.J., Brushart, T.M., 1996. Reinnervation accuracy of the rat femoral nerve by motor and sensory neurons. Journal of Neuroscience 16 (18), 5698–5703.
Madison, R.D., Da Silva, C.F., Dikkes, P., 1988. Entubulation repair with protein additives increases the maximum nerve gap distance successfully bridged with tubular prostheses. Brain Research 447 (2), 325–334.
Madison, R.D., Robinson, G.A., Chadaram, S.R., 2007. The specificity of motor neurone regeneration (preferential reinnervation). Acta Physiologica (Oxford) 189 (2), 201–206.
Mandolesi, G., Madeddu, F., Bozzi, Y., Maffei, L., Ratto, G.M., 2004. Acute physiological response of mammalian central neurons to axotomy: ionic regulation and electrical activity. FASEB Journal 18 (15), 1934–1936.
Mao, Y., Schwarzbauer, J.E., 2005. Stimulatory effects of a three-dimensional microenvironment on cell-mediated fibronectin fibrillogenesis. Journal of Cell Science 118 (Pt 19), 4427–4436.
Marchionni, M.A., Goodearl, A.D., Chen, M.S., Bermingham-McDonogh, O., Kirk, C., Hendricks, M., Danehy, F., Misumi, D., Sudhalter, J., Kobayashi, K., 1993. Glial growth factors are alternatively spliced erbB2 ligands expressed in the nervous system. Nature 362 (6418), 312–318.
Marino, P., Norreel, J.C., Schachner, M., Rougon, G., Amoureux, M.C., 2009. A polysialic acid mimetic peptide promotes functional recovery in a mouse model of spinal cord injury. Experimental Neurolgy 219 (1), 163–174.
Martini, R., Schachner, M., Brushart, T.M., 1994. The L2/HNK-1 carbohydrate is preferentially expressed by previously motor axon-associated Schwann cells in reinnervated peripheral nerves. Journal of Neuroscience 14 (11 Pt 2), 7180–7191.
Martini, R., Xin, Y., Schmitz, B., Schachner, M., 1992. The L2/HNK-1 carbohydrate epitope is involved in the preferential outgrowth of motor neurons on ventral roots and motor nerves. European Journal of Neuroscience 4 (7), 628–639.
Mason, M.R., Lieberman, A.R., Anderson, P.N., 2003. Corticospinal neurons up-regulate a range of growth-associated genes following intracortical, but not spinal, axotomy. European Journal of Neuroscience 18 (4), 789–802.
Mason, M.R., Tannemaat, M.R., Malessy, M.J., Verhaagen, J., 2011. Gene therapy for the peripheral nervous system: a strategy to repair the injured nerve? Current Gene Therapy 11 (2), 75–89.
Matheson, C.R., Wang, J., Collins, F.D., Yan, Q., 1997. Long-term survival effects of GDNF on neonatal rat facial motoneurons after axotomy. Neuroreport 8 (7), 1739–1742.
Mathews, G.A., Ffrench-Constant, C., 1995. Embryonic fibronectins are up-regulated following peripheral nerve injury in rats. Journal of Neurobiology 26 (2), 171–188.
McKeehan, W.L., Wang, F., Kan, M., 1998. The heparan sulfate-fibroblast growth factor family: diversity of structure and function. Progress in Nucleic Acid Research and Molecular Biology 59, 135–176.
McQuarrie, I.G., Brady, S.T., Lasek, R.J., 1989. Retardation in the slow axonal transport of cytoskeletal elements during maturation and aging. Neurobiology of Aging 10 (4), 359–365.
McQuarrie, I.G., Grafstein, B., Gershon, M.D., 1977. Axonal regeneration in the rat sciatic nerve: effect of a conditioning lesion and of dbcAMP. Brain Research 132 (3), 443–453.
Mears, S., Schachner, M., Brushart, T.M., 2003. Antibodies to myelin-associated glycoprotein accelerate preferential motor reinnervation. Journal of the Peripheral Nervous System 8 (2), 91–99.
Mehanna, A., Mishra, B., Kurschat, N., Schulze, C., Bian, S., Loers, G., Irintchev, A., Schachner, M., 2009. Polysialic acid glycomimetics promote myelination and functional recovery after peripheral nerve injury in mice. Brain 132 (Pt 6), 1449–1462.
Meisinger, C., Zeschnigk, C., Grothe, C., 1996. In vivo and in vitro effect of glucocorticoids on fibroblast growth factor (FGF)-2 and FGF receptor 1 expression. Journal of Biological Chemistry 271 (28), 16520–16525.
Mi, R., Chen, W., Hoke, A., 2007. Pleiotrophin is a neurotrophic factor for spinal motor neurons. Proceedings of the National Academy of Sciences of the United States of America 104 (11), 4664–4669.
Michaelevski, I., Medzihradszky, K.F., Lynn, A., Burlingame, A.L., Fainzilber, M., 2010. Axonal transport proteomics reveals mobilization of translation machinery to

the lesion site in injured sciatic nerve. Molecular and Cellular Proteomics 9 (5), 976–987.
Michalski, B., Bain, J.R., Fahnestock, M., 2008. Long-term changes in neurotrophic factor expression in distal nerve stump following denervation and reinnervation with motor or sensory nerve. Journal of Neurochemistry 105 (4), 1244–1252.
Mills, C.D., Allchorne, A.J., Griffin, R.S., Woolf, C.J., Costigan, M., 2007. GDNF selectively promotes regeneration of injury-primed sensory neurons in the lesioned spinal cord. Molecular and Cellular Neuroscience 36 (2), 185–194.
Mineo, C., James, G.L., Smart, E.J., Anderson, R.G., 1996. Localization of epidermal growth factor-stimulated Ras/Raf-1 interaction to caveolae membrane. Journal of Biological Chemistry 271 (20), 11930–11935.
Miner, J.H., 2008. Laminins and their roles in mammals. Microscopy Research and Technique 71 (5), 349–356.
Mohiuddin, L., Delcroix, J.D., Fernyhough, P., DR, T., 1999. Focally administered nerve growth factor suppresses molecular regenerative responses of axotomized peripheral afferents in rats. Neuroscience 91 (1), 265–271.
Molander, C., Aldskogius, H., 1992. Directional specificity of regenerating primary sensory neurons after peripheral nerve crush or transection and epineurial suture A sequential double-labeling study in the rat. Restorative Neurology and Neuroscience 4 (5), 339–344.
Molliver, D.C., Wright, D.E., Leitner, M.L., Parsadanian, A.S., Doster, K., Wen, D., Yan, Q., Snider, W.D., 1997. IB4-binding DRG neurons switch from NGF to GDNF dependence in early postnatal life. Neuron 19 (4), 849–861.
Moradzadeh, A., Borschel, G.H., Luciano, J.P., Whitlock, E.L., Hayashi, A., Hunter, D.A., Mackinnon, S.E., 2008. The impact of motor and sensory nerve architecture on nerve regeneration. Experimental Neurology 212 (2), 370–376.
Mortimer, D., Fothergill, T., Pujic, Z., Richards, L.J., Goodhill, G.J., 2008. Growth cone chemotaxis. Trends in Neurosciences 31 (2), 90–98.
Mu, X., Silos-Santiago, I., Carroll, S.L., Snider, W.D., 1993. Neurotrophin receptor genes are expressed in distinct patterns in developing dorsal root ganglia. Journal of Neuroscience 13 (9), 4029–4041.
Mueller, B.K., 1999. Growth cone guidance: first steps towards a deeper understanding. Annual Review of Neuroscience 22, 351–388.
Mukhopadhyay, G., Doherty, P., Walsh, F.S., Crocker, P.R., Filbin, M.T., 1994. A novel role for myelin-associated glycoprotein as an inhibitor of axonal regeneration. Neuron 13 (3), 757–767.
Nadeau, S., Hein, P., Fernandes, K.J., Peterson, A.C., Miller, F.D., 2005. A transcriptional role for C/EBP beta in the neuronal response to axonal injury. Molecular and Cellular Neuroscience 29 (4), 525–535.
Navarro, X., Rodriguez, F.J., Ceballos, D., Verdu, E., 2003. Engineering an artificial nerve graft for the repair of severe nerve injuries. Medical and Biological Engineering and Computing 41 (2), 220–226.
Navarro, X., Rodriguez, F.J., Labrador, R.O., Buti, M., Ceballos, D., Gomez, N., Cuadras, J., Perego, G., 1996. Peripheral nerve regeneration through bioresorbable and durable nerve guides. Journal of the Peripheral Nervous System 1 (1), 53–64.
Navarro, X., Vivó, M., Valero-Cabré, A., 2007. Neural plasticity after peripheral nerve injury and regeneration. Progress in Neurobiology 82 (4), 163–201.
Nichols, C.M., Brenner, M.J., Fox, I.K., Tung, T.H., Hunter, D.A., Rickman, S.R., Mackinnon, S.E., 2004. Effects of motor versus sensory nerve grafts on peripheral nerve regeneration. Experimental Neurology 190 (2), 347–355.
Niethammer, P., Delling, M., Sytnyk, V., Dityatev, A., Fukami, K., Schachner, M., 2002. Cosignaling of NCAM via lipid rafts and the FGF receptor is required for neuritogenesis. Journal of Cell Biology 157 (3), 521–532.
Noguchi, K., Senba, E., Morita, Y., Sato, M., Tohyama, M., 1990. Co-expression of alpha-CGRP and beta-CGRP mRNAs in the rat dorsal root ganglion cells. Neuroscience Letters 108 (1–2), 1–5.
Ornitz, D.M., 2000. FGFs, heparan sulfate and FGFRs: complex interactions essential for development. Bioessays 22 (2), 108–112.
Ornitz, D.M., Itoh, N., 2001. Fibroblast growth factors. Genome Biology 2 (3), 3005 (Reviews).
Ornitz, D.M., Xu, J., Colvin, J.S., McEwen, D.G., MacArthur, C.A., Coulier, F., Gao, G., Goldfarb, M., 1996. Receptor specificity of the fibroblast growth factor family. Journal of Biological Chemistry 271 (25), 15292–15297.
Palazzo, A.F., Eng, C.H., Schlaepfer, D.D., Marcantonio, E.E., Gundersen, G.G., 2004. Localized stabilization of microtubules by integrin- and FAK-facilitated Rho signaling. Science 303 (5659), 836–839.
Pankov, R., Yamada, K.M., 2002. Fibronectin at a glance. Journal of Cell Science 115 (Pt 20), 3861–3863.
Paratcha, G., Ledda, F., Baars, L., Coulpier, M., Besset, V., Anders, J., Scott, R., Ibáñez, C.F., 2001. Released GFRalpha1 potentiates downstream signaling, neuronal survival, and differentiation via a novel mechanism of recruitment of c-Ret to lipid rafts. Neuron 29 (1), 171–184.
Park, K.K., Liu, K., Hu, Y., Smith, P.D., Wang, C., Cai, B., Xu, B., Connolly, L., Kramvis, I., Sahin, M., He, Z., 2008. Promoting axon regeneration in the adult CNS by modulation of the PTEN/mTOR pathway. Science 322 (5903), 963–966.
Parkinson, D.B., Bhaskaran, A., Arthur-Farraj, P., Noon, L.A., Woodhoo, A., Lloyd, A.C., Feltri, M.L., Wrabetz, L., Behrens, A., Mirsky, R., Jessen, K.R., 2008. c-Jun is a negative regulator of myelination. Journal of Cell Biology 181 (4), 625–637.
Peiró, S., Comella, J.X., Enrich, C., Martín-Zanca, D., Rocamora, N., 2000. PC12 cells have caveolae that contain TrkA. Caveolae-disrupting drugs inhibit nerve growth factor-induced, but not epidermal growth factor-induced, MAPK phosphorylation. Journal of Biological Chemistry 275 (48), 37846–37852.
Peng, H.B., Ali, A.A., Dai, Z., Daggett, D.F., Raulo, E., Rauvala, H., 1995. The role of heparin-binding growth-associated molecule (HB-GAM) in the postsynaptic induction in cultured muscle cells. Journal of Neuroscience 15 (4), 3027–3038.
Perlson, E., Hanz, S., Ben-Yaakov, K., Segal-Ruder, Y., Seger, R., Fainzilber, M., 2005. Vimentin-dependent spatial translocation of an activated MAP kinase in injured nerve. Neuron 45 (5), 715–726.
Pertz, O.C., Wang, Y., Yang, F., Wang, W., Gay, L.J., Gristenko, M.A., Clauss, T.R., Anderson, D.J., Liu, T., Auberry, K.J., Camp, D.G., Smith, R.D., Klemke, R.L., 2008. Spatial mapping of the neurite and soma proteomes reveals a functional Cdc42/Rac regulatory network. Proceedings of the National Academy of Sciences of the United States of America 105 (6), 1931–1936.
Piehl, F., Frisen, J., Risling, M., Hokfelt, T., Cullheim, S., 1994. Increased trkB mRNA expression by axotomized motoneurones. Neuroreport 5 (6), 697–700.
Pitto, M., Mutoh, T., Kuriyama, M., Ferraretto, A., Palestini, P., Masserini, M., 1998. Influence of endogenous GM1 ganglioside on TrkB activity, in cultured neurons. FEBS Letters 439 (1-2), 93–96.
Plantman, S., Patarroyo, M., Fried, K., Domogatskaya, A., Tryggvason, K., Hammarberg, H., Cullheim, S., 2008. Integrin-laminin interactions controlling neurite outgrowth from adult DRG neurons in vitro. Molecular and Cellular Neuroscience 39 (1), 50–62.
Plunet, W., Kwon, B.K., Tetzlaff, W., 2002. Promoting axonal regeneration in the central nervous system by enhancing the cell body response to axotomy. Journal of Neuroscience Research 68 (1), 1–6.
Politis, M.J., Ederle, K., Spencer, P.S., 1982. Tropism in nerve regeneration in vivo. Attraction of regenerating axons by diffusible factors derived from cells in distal nerve stumps of transected peripheral nerves. Brain Research 253 (1-2), 1–12.
Politis, M.J., Spencer, P.S., 1981. A method to separate spatially the temporal sequence of axon-Schwann cell interaction during nerve regeneration. Journal of Neurocytology 10 (2), 221–232.
Poteryaev, D., Titievsky, A., Sun, Y.F., Thomas-Crusells, J., Lindahl, M., Billaud, M., Arumäe, U., Saarma, M., 1999. GDNF triggers a novel ret-independent Src kinase family-coupled signaling via a GPI-linked GDNF receptor alpha1. FEBS Letters 463 (1-2), 63–66.
Pu, S.F., Zhuang, H.X., Marsh, D.J., Ishii, D.N., 1999. Insulin-like growth factor-II increases and IGF is required for postnatal rat spinal motoneuron survival following sciatic nerve axotomy. Journal of Neuroscience Research 55 (1), 9–16.
Rabinovsky, E.D., Smith, G.M., Browder, D.P., Shine, H.D., McManaman, J.L., 1992. Peripheral-nerve injury down-regulates CNTF expression in adult-rat sciatic-nerves. Journal of Neuroscience Research 31 (1), 188–192.
Raivich, G., Bohatschek, M., Da Costa, C., Iwata, O., Galiano, M., Hristova, M., Nateri, A.S., Makwana, M., Riera-Sans, L., Wolfer, D.P., Lipp, H.P., Aguzzi, A., Wagner, E.F., Behrens, A., 2004. The AP-1 transcription factor c-Jun is required for efficient axonal regeneration. Neuron 43 (1), 57–67.
Raivich, G., Dumoulin, F.L., Streit, W.J., Kreutzberg, G.W., 1992. Calcitonin gene-related peptide (CGRP) in the regenerating rat sciatic nerve. Restorative Neurology and Neuroscience 4 (2), 107–115.
Raivich, G., Hellweg, R., Kreutzberg, G.W., 1991. NGF receptor-mediated reduction in axonal NGF uptake and retrigrade transport following sciatic nerve injury and during regeneration. Neuron 7 (1), 151–164.
Raivich, G., Kreutzberg, G.W., 1994. Pathophysiology of glial growth factor receptors. Glia 11 (2), 129–146.
Raivich, G., Makwana, M., 2007. The making of successful axonal regeneration: genes, molecules and signal transduction pathways. Brain Research Reviews 53 (2), 287–311.
Rankin, S.L., Guy, C.S., Rahimtula, M., Mearow, K.M., 2008. Neurotrophin-induced upregulation of p75NTR via a protein kinase C-delta-dependent mechanism. Brain Research 1217, 10–24.
Ranscht, B., 2000. Cadherins: molecular codes for axon guidance and synapse formation. International Journal of Developmental Neuroscience 18 (7), 643–651.
Rao, M.S., Sun, Y., Escary, J.L., Perreau, J., Tresser, S., Patterson, P.H., Zigmond, R.E., Brulet, P., Landis, S.C., 1993. Leukemia inhibitory factor mediates an injury response but not a target-directed developmental transmitter switch in sympathetic neurons. Neuron 11 (6), 1175–1185.
Redett, R., Jari, R., Crawford, T., Chen, Y.G., Rohde, C., Brushart, T.M., 2005. Peripheral pathways regulate motoneuron collateral dynamics. Journal of Neuroscience 25 (41), 9406–9412.
Rich, K.M., Luszczynski, J.R., Osborne, P.A., Johnson Jr., E.M., 1987. Nerve growth factor protects adult sensory neurons from cell death and atrophy caused by nerve injury. Journal of Neurocytology 16 (2), 261–268.
Rishal, I., Fainzilber, M., 2010. Retrograde signaling in axonal regeneration. Experimental Neurology 223 (1), 5–10.
Rodriguez, F.J., Verdu, E., Ceballos, D., Navarro, X., 2000. Nerve guides seeded with autologous schwann cells improve nerve regeneration. Experimental Neurology 161 (2), 571–584.
Rossi, F., Gianola, S., Corvetti, L., 2007. Regulation of intrinsic neuronal properties for axon growth and regeneration. Progress in Neurobiology 81 (1), 1–28.
Saffell, J.L., Williams, E.J., Mason, I.J., Walsh, F.S., Doherty, P., 1997. Expression of a dominant negative FGF receptor inhibits axonal growth and FGF receptor phosphorylation stimulated by CAMs. Neuron 18 (2), 231–242.
Saito, H., Nakao, Y., Takayama, S., Toyama, Y., Asou, H., 2005. Specific expression of an HNK-1 carbohydrate epitope and NCAM on femoral nerve Schwann cells in mice. Neuroscience Research 53 (3), 314–322.
Schéele, S., Nyström, A., Durbeej, M., Talts, J.F., Ekblom, M., Ekblom, P., 2007. Laminin isoforms in development and disease. Journal of Molecular Medicine 85 (8), 825–836.
Scherer, S.S., Wang, D.Y., Kuhn, R., Lemke, G., Wrabetz, L., Kamholz, J., 1994. Axons regulate Schwann cell expression of the POU transcription factor SCIP. Journal of Neuroscience 14 (4), 1930–1942.

Scherer, S.S., Salzer, J.L., 1996. Axon-Schwann cell interactions in peripheral nerve regeneration. In: Jessen, K.R., Richardson, W.D. (Eds.), Glial Cell Development. Bios Scientific, Oxford.

Schmidt, H., Werner, M., Heppenstall, P.A., Henning, M., Moré, M.I., Kühbandner, S., Lewin, G.R., Hofmann, F., Feil, R., Rathjen, F.G., 2002. cGMP-mediated signaling via cGKIalpha is required for the guidance and connectivity of sensory axons. Journal of Cell Biology 159 (3), 489–498.

Schneider, V.A., Granato, M., 2003. Motor axon migration: a long way to go. Developmental Biology 263 (1), 1–11.

Schulze, E., Asai, D.J., Bulinski, C.J., Kirschner, M., 1987. Post-translational modification and microtubule stability. Journal of Cell Biology 105 (5), 2167–2177.

Schwaiger, F.W., Hager, G., Schmitt, A.B., Horvat, A., Hager, G., Streif, R., Spitzer, C., Gamal, S., Breuer, S., Brook, G.A., Nacimiento, W., Kreutzberg, G.W., 2000. Peripheral but not central axotomy induces changes in Janus kinases (JAK) and signal transducers and activators of transcription (STAT). European Journal of Neuroscience 12 (4), 1165–1176.

Schwartz, M.A., Ginsberg, M.H., 2002. Networks and crosstalk: integrin signalling spreads. Nature Cell Biology 4 (4), E65–E68.

Sendtner, M., Kreutzberg, G.W., Thoenen, H., 1990. Ciliary neurotrophic factor prevents the degeneration of motor neurons after axotomy. Nature 345 (6274), 440–441.

Shadiack, A.M., Sun, Y., Zigmond, R.E., 2001. Nerve growth factor antiserum induces axotomy-like changes in neuropeptide expression in intact sympathetic and sensory neurons. Journal of Neuroscience 21 (2), 363–371.

Shen, H., Chung, J.M., Coggeshall, R.E., Chung, K., 1999. Changes in trkA expression in the dorsal root ganglion after peripheral nerve injury. Experimental Brain Research 127 (2), 141–146.

Shen, Y.J., DeBellard, M.E., Salzer, J.L., Roder, J., Filbin, M.T., 1998. Myelin-associated glycoprotein in myelin and expressed by Schwann cells inhibits axonal regeneration and branching. Molecular and Cellular Neuroscience 12 (1–2), 79–91.

Siegel, S.G., Patton, B., English, A.W., 2000. Ciliary neurotrophic factor is required for motoneuron sprouting. Experimental Neurology 166 (2), 205–212.

Simon, M., Porter, R., Brown, R., Coulton, G.R., Terenghi, G., 2003. Effect of NT-4 and BDNF delivery to damaged sciatic nerves on phenotypic recovery of fast and slow muscles fibres. European Journal of Neuroscience 18 (9), 2460–2466.

Simons, K., Toomre, D., 2000. Lipid rafts and signal transduction. Nature Reviews Molecular Cell Biology 1 (1), 31–39.

Skene, J., 1989. Axonal growth-associated proteins. Annual Review of Neuroscience 12, 127–156.

Skene, J.H., Willard, M., 1981. Axonally transported proteins associated with axon growth in rabbit central and peripheral nervous systems. Journal of Cell Biology 89 (1), 96–103.

Smith, D.S., Skene, J.H., 1997. A transcription-dependent switch controls competence of adult neurons for distinct modes of axon growth. Journal of Neuroscience 17 (2), 646–658.

Song, H.J., Ming, G.L., Poo, M.M., 1997. cAMP-induced switching in turning direction of nerve growth cones. Nature 388 (6639), 275–279.

Song, H.J., Poo, M.M., 1999. Signal transduction underlying growth cone guidance by diffusible factors. Current Opinion in Neurobiology 9 (3), 355–363.

Steinthorsdottir, V., Stefansson, H., Ghosh, S., Birgisdottir, B., Bjornsdottir, S., Fasquel, A.C., Olafsson, O., Stefansson, K., Gulcher, J.R., 2004. Multiple novel transcription initiation sites for NRG1. Gene 342 (1), 97–105.

Sterne, G.D., Brown, R.A., Green, C.J., Terenghi, G., 1997a. Neurotrophin-3 delivered locally via fibronectin mats enhances peripheral nerve regeneration. European Journal of Neuroscience 9 (7), 1388–1396.

Sterne, G.D., Coulton, G.R., Brown, R.A., Green, C.J., Terenghi, G., 1997b. Neurotrophin-3-enhanced nerve regeneration selectively improves recovery of muscle fibers expressing myosin heavy chains 2b. Journal of Cell Biology 139 (3), 709–715.

Stoll, G., Muller, H.W., 1999. Nerve injury, axonal degeneration and neural regeneration: basic insights. Brain Pathology 9 (2), 313–325.

Streit, W.J., Hurley, S.D., McGraw, T.S., Semple-Rowland, S.L., 2000. Comparative evaluation of cytokine profiles and reactive gliosis supports a critical role for interleukin-6 in neuron-glia signaling during regeneration. Journal of Neuroscience Research 61 (1), 10–20.

Sulaiman, O.A.R., Gordon, T., 2000. Effects of short- and long-term Schwann cell denervation on peripheral nerve regeneration, myelination, and size. Glia 32 (3), 234–246.

Suzuki, S., Numakawa, T., Shimazu, K., Koshimizu, H., Hara, T., Hatanaka, H., Mei, L., Lu, B., Kojima, M., 2004. BDNF-induced recruitment of TrkB receptor into neuronal lipid rafts: roles in synaptic modulation. Journal of Cell Biology 167 (6), 1205–1215.

Szebenyi, G., Fallon, J.F., 1999. Fibroblast growth factors as multifunctional signaling factors. International Review of Cytology 185, 45–106.

Takada, Y., Ye, X., Simon, S., 2007. The integrins. Genome Biology 8, 215.

Tang, B.L., 2003. Inhibitors of neuronal regeneration: mediators and signaling mechanisms. Neurochemistry International 42 (3), 189–203.

Taniuchi, M., Clark, H.B., Schweitzer, J.B., Johnson Jr., E.M., 1988. Expression of nerve growth factor receptors by Schwann cells of axotomized peripheral nerves: ultrastructural location, suppression by axonal contact, and binding properties. Journal of Neuroscience 8 (2), 664–681.

Tannemaat, M.R., Eggers, R., Hendriks, W.T., de Ruiter, G.C., van Heerikhuize, J.J., Pool, C.W., Malessy, M.J., Boer, G.J., Verhaagen, J., 2008. Differential effects of lentiviral vector-mediated overexpression of nerve growth factor and glial cell line-derived neurotrophic factor on regenerating sensory and motor axons in the transected peripheral nerve. European Journal of Neuroscience 28 (8), 1467–1479.

Tansey, M.G., Baloh, R.H., Milbrandt, J., Johnson Jr., E.M., 2000. GFRalpha-mediated localization of RET to lipid rafts is required for effective downstream signaling, differentiation, and neuronal survival. Neuron 25 (3), 611–623.

Terenghi, G., 1999. Peripheral nerve regeneration and neurotrophic factors. Journal of Anatomy 194 (Pt 1), 1–14.

Tessier-Lavigne, M., Goodman, C.S., 1996. The molecular biology of axon guidance. Science 274 (5290), 1123–1133.

Tham, S., Dowsing, B., Finkelstein, D., Donato, R., Cheema, S.S., Bartlett, P.F., Morrison, W.A., 1997. Leukemia inhibitory factor enhances the regeneration of transected rat sciatic nerve and the function of reinnervated muscle. Journal of Neuroscience Research 47 (2), 208–215.

Thompson, S.W., Vernallis, A.B., Heath, J.K., Priestley, J.V., 1997. Leukaemia inhibitory factor is retrogradely transported by a distinct population of adult rat sensory neurons: co-localization with trkA and other neurochemical markers. European Journal of Neuroscience 9 (6), 1244–1251.

Toth, C.C., Willis, D., Twiss, J.L., Walsh, S., Martinez, J.A., Liu, W.Q., Midha, R., Zochodne, D.W., 2009. Locally synthesized calcitonin gene-related peptide has a critical role in peripheral nerve regeneration. Journal of Neuropathology and Experimental Neurology 68 (3), 326–337.

Treanor, J.J., Goodman, L., de Sauvage, F., Stone, D.M., Poulsen, K.T., Beck, C.D., Gray, C., Armanini, M.P., Pollock, R.A., Hefti, F., Phillips, H.S., Goddard, A., Moore, M.W., Buj-Bello, A., Davies, A.M., Asai, N., Takahashi, M., Vandlen, R., Henderson, C.E., Rosenthal, A., 1996. Characterization of a multicomponent receptor for GDNF. Nature 382 (6586), 80–83.

Trupp, M., Arenas, E., Fainzilber, M., Nilsson, A., Sieber, S., Grigoriou, B.A., Kilkenny, M., Salazar-Grueso, C., Pachnis, E., Arumäe, V.U., 1996. Functional receptor for GDNF encoded by the c-ret proto-oncogene. Nature 381 (6585), 785–789.

Trupp, M., Belluardo, N., Funakoshi, H., Ibáñez, C.F., 1997. Complementary and overlapping expression of glial cell line-derived neurotrophic factor (GDNF), c-ret proto-oncogene, and GDNF receptor-alpha indicates multiple mechanisms of trophic actions in the adult rat CNS. Journal of Neuroscience 17 (10), 3554–3567.

Trupp, M., Ryden, M., Jornvall, H., Funakoshi, H., Timmusk, T., Arenas, E., Ibanez, C.F., 1995. Peripheral expression and biological activities of GDNF, a new neurotrophic factor for avian and mammalian peripheral neurons. Journal of Cell Biology 130 (1), 137–148.

Trupp, M., Scott, R., Whittemore, S.R., Ibanez, C.F., 1999. Ret-dependent and -independent mechanisms of glial cell line-derived neurotrophic factor signaling in neuronal cells. Journal of Biological Chemistry 274 (30), 20885–20894.

Tsui-Pierchala, B.A., Encinas, M., Milbrandt, J., Johnson, E.M.J., 2002. Lipid rafts in neuronal signaling and function. Trends in Neurosciences 25 (8), 412–417.

Tsujino, H., Kondo, E., Fukuoka, T., Dai, Y., Tokunaga, A., Miki, K., Yonenobu, K., Ochi, T., Noguchi, K., 2000. Activating transcription factor 3 (ATF3) induction by axotomy in sensory and motoneurons: a novel neuronal marker of nerve injury. Molecular and Cellular Neuroscience 15 (2), 170–182.

Tzu, J., Marinkovich, M.P., 2008. Bridging structure with function: structural, regulatory, and developmental role of laminins. International Journal of Biochemistry and Cell Biology 40 (2), 199–214.

Udina, E., Ladak, A., Furey, M., Brushart, T., Tyreman, N., Gordon, T., 2010. Rolipram-induced elevation of cAMP or chondroitinase ABC breakdown of inhibitory proteoglycans in the extracellular matrix promotes peripheral nerve regeneration. Experimental Neurology 223 (1), 143–152.

Ulenkate, H.J.L.M., Kaal, E.C.A., Gispen, W.H., Jennekens, F.G.I., 1994. Ciliary neurotrophic factor improves muscle fibre reinnervation after facial nerve crush in young rats. Acta Neuropathologica (Berl.) 88 (6), 558–564.

Valero-Cabre, A., Navarro, X., 2002. Functional impact of axonal misdirection after peripheral nerve injuries followed by graft or tube repair. Journal of Neurotrauma 19 (11), 1475–1485.

Valero-Cabre, A., Tsironis, K., Skouras, E., Navarro, X., Neiss, W.F., 2004. Peripheral and spinal motor reorganization after nerve injury and repair. Journal of Neurotrauma 21 (1), 95–108.

Van der Flier, A., Sonnenberg, A., 2001. Function and interactions of integrins. Cell and Tissue Research 305 (3), 285–298.

Verdu, E., Labrador, R.O., Rodriguez, F.J., Ceballos, D., Fores, J., Navarro, X., 2002. Alignment of collagen and laminin-containing gels improve nerve regeneration within silicone tubes. Restorative Neurology and Neuroscience 20 (5), 169–179.

Verma, P., Chierzi, S., Codd, A.M., Campbell, D.S., Meyery, R.L., Holt, C.E., Fawcett, J.W., 2005. Axonal protein synthesis and degradation are necessary for efficient growth cone regeneration. Journal of Neuroscience 25 (2), 331–342.

Villar, M.J., Cortés, R., Theodorsson, E., Wiesenfeld-Hallin, Z., Schalling, M., Fahrenkburg, J., Emson, P.C., Hökfelt, T., 1989. Neuropeptide expression in rat dorsal root ganglion cells and spinal cord after peripheral nerve injury with special reference to galanin. Neuroscience 33 (3), 587–604.

Villar, M.J., Wiesenfeld-Hallin, Z., Xu, X.J., Theodorsson, E., Emson, P.C., Hökfelt, T., 1991. Further studies on galanin-, substance P-, and CGRP-like immunoreactivities in primary sensory neurons and spinal cord: effects of dorsal rhizotomies and sciatic nerve lesions. Experimental Neurology 112 (1), 29–39.

Vogelezang, M.G., Liu, Z., Relvas, J.B., Raivich, G., Scherer, S.S., ffrench-Constant, C., 2001. Alpha4 integrin is expressed during peripheral nerve regeneration and enhances neurite outgrowth. Journal of Neuroscience 21 (17), 6732–6744.

Vogelezang, M.G., Scherer, S.S., Fawcett, J.W., ffrench-Constant, C., 1999. Regulation of fibronectin alternative splicing during peripheral nerve repair. Journal of Neuroscience Research 56 (4), 323–333.

Vögelin, E., Baker, J.M., Gates, J., Dixit, V., Constantinescu, M.A., Jones, N.F., 2006. Effects of local continuous release of brain derived neurotrophic factor (BDNF) on peripheral nerve regeneration in a rat model. Experimental Neurology 199 (2), 348–353.

Vrbova, G., Mehra, N., Shanmuganathan, H., Tyreman, N., Schachner, M., Gordon, T., 2009. Chemical communication between regenerating motor axons and Schwann cells in the growth pathway. European Journal of Neuroscience 30 (3), 366–375.

Wakisaka, S., Kajander, K.C., Bennett, G.J., 1991. Increased neuropeptide (NPY)-like immunoreactivity in rat sensory neurons following peripheral axotomy. Neuroscience Letters 124 (2), 200–203.

Wallquist, W., Patarroyo, M., Thams, S., Carlstedt, T., Stark, B., Cullheim, S., Hammarberg, H., 2002. Laminin chains in rat and human peripheral nerve: distribution and regulation during development and after axonal injury. Journal of Comparative Neurology 454 (3), 284–293.

Wallquist, W., Plantman, S., Thams, S., Thyboll, J., Kortesmaa, J., Lännergren, J., Domogatskaya, A., Ogren, S.O., Risling, M., Hammarberg, H., Tryggvason, K., Cullheim, S., 2005. Impeded interaction between Schwann cells and axons in the absence of laminin alpha4. Journal of Neuroscience 25 (14), 3692–3700.

Wallquist, W., Zelano, J., Plantman, S., Kaufman, S.J., Cullheim, S., Hammarberg, H., 2004. Dorsal root ganglion neurons up-regulate the expression of laminin-associated integrins after peripheral but not central axotomy. Journal of Comparative Neurology 480 (2), 162–169.

Wang, G.Y., Hirai, K., Shimada, H., 1992. The role of laminin, a component of Schwann cell basal lamina, in rat sciatic nerve regeneration within antiserum-treated nerve grafts. Brain Research 570 (1-2), 116–125.

Webber, C.A., Xu, Y., Vanneste, K.J., Martinez, J.A., Verge, V.M.K., Zochodne, D.W., 2008. Guiding adult mammalian sensory axons during regeneration. Journal of Neuropathology and Experimental Neurology 67 (3), 212–222.

Wen, D., Peles, E., Cupples, R., Suggs, S.V., Bacus, S.S., Luo, Y., Trail, G., Hu, S., Silbiger, S.M., Levy, R.B., 1992. Neu differentiation factor: a transmembrane glycoprotein containing an EGF domain and an immunoglobulin homology unit. Cell 69 (3), 559–572.

Wen, D., Suggs, S.V., Karunagaran, D., Liu, N., Cupples, R.L., Luo, Y., Janssen, A.M., Ben-Baruch, N., Trollinger, D.B., Jacobsen, V.L., 1994. Structural and functional aspects of the multiplicity of Neu differentiation factors. Molecular and Cellular Biology 14 (3), 1909–1919.

Wen, Z., Zheng, J.Q., 2006. Directional guidance of nerve growth cones. Current Opinion in Neurobiology 16 (1), 52–58.

Werner, A., Willem, M., Jones, L.L., Kreutzberg, G.W., Mayer, U., Raivich, G., 2000. Impaired axonal regeneration in alpha7 integrin-deficient mice. Journal of Neuroscience 20 (5), 1822–1830.

Williams, L.R., 1987. Exogenous fibrin matrix precursors stimulate the temporal progress of nerve regeneration within a silicone chamber. Neurochemical Research 12 (10), 851–860.

Williams, L.R., Longo, F.M., Powell, H.C., Lundborg, G., Varon, S., 1983. Spatial-temporal progress of peripheral nerve regeneration within a silicone chamber. Parameters for a biossay. Journal of Comparative Neurology 218 (4), 160–170.

Willis, D.E., Twiss, J.L., 2005. The evolving roles of axonally synthesized proteins in regeneration. Current Opinion in Neurobiology 16 (1), 111–118.

Witzel, C., Rohde, C., Brushart, T.M., 2005. Pathway sampling by regenerating peripheral axons. Journal of Comparative Neurology 485 (3), 183–190.

Wong, B.J., Mattox, D.E., 1991. The effects of polyamines and polyamine inhibitors on rat sciatic and facial nerve regeneration. Experimental Neurology 111 (2), 263–266.

Wu, C., Butz, S., Ying, Y., Anderson, R.G., 1997. Tyrosine kinase receptors concentrated in caveolae-like domains from neuronal plasma membrane. Journal of Biological Chemistry 272 (6), 3554–3559.

Yamada, K.M., Even-Ram, S., 2002. Integrin regulation of growth factor receptors. Nature Cell Biology 4 (4), E75–E76.

Yamamoto, M., Toya, Y., Schwencke, C., Lisanti, M.P., Myers, M.G.J., Ishikawa, Y., 1998. Caveolin is an activator of insulin receptor signaling. Journal of Biological Chemistry 273 (41), 26962–26968.

Yan, Q., Matheson, C., Lopez, O.T., 1995. In vivo neurotrophic effects of GDNF on neonatal and adult facial motor neurons. Nature 373 (6512), 341–344.

Yildiz, M., Karlidag, T., Yalcin, S., Ozogul, C., Keles, E., Alpay, H.C., Yanilmaz, M., 2011. Efficacy of glial growth factor and nerve growth factor on the recovery of traumatic facial paralysis. European Archives of Otorhinolaryngology 268 (8), 1127–1133.

Zhang, J.Y., Luo, X.G., Xian, C.J., Liu, Z.H., Zhou, X.F., 2000. Endogenous BDNF is required for myelination and regeneration of injured sciatic nerve in rodents. European Journal of Neuroscience 12 (12), 4171–4180.

Zhang, X., Verge, V.M., Wiesenfeld-Hallin, Z., Piehl, F., Hökfelt, T., 1993. Expression of neuropeptides and neuropeptide mRNAs in spinal cord after axotomy in the rat, with special reference to motoneurons and galanin. Experimental Brain Research 93 (3), 450–461.

Zheng, L.F., Wang, R., Xu, Y.Z., Yi, X.N., Zhang, J.W., Zeng, Z.C., 2008. Calcitonin gene-related peptide dynamics inrat dorsal root ganglia and spinal cord following different sciatic nerve injuries. Brain Research 1187, 20–32.

Zhao, Z., Alam, S., Oppenheim, R.W., Prevette, D.M., Evenson, A., Parsadanian, A., 2004. Overexpression of glial cell line-derived neurotrophic factor in the CNS rescues motoneurons from programmed cell death and promotes their long-term survival following axotomy. Experimental Neurology 190 (2), 356–372.

Zhou, T.H., Ling, K., Guo, J., Zhou, H., Wu, Y.L., Jing, Q., Ma, L., Pei, G., 2000. Identification of a human brain-specific isoform of mammalian STE20-like kinase 3 that is regulated by cAMP-dependent protein kinase. Journal of Biological Chemistry 275 (4), 2513–2519.

Zhou, X.F., Rush, R.A., 1995. Peripheral projections of rat primary sensory neurons immunoreactive for neurotrophin-3. Journal of Comparative Neurology 363 (1), 69–77.

Zuo, J., Hernandez, Y.J., Muir, D., 1998. Chondroitin sulfate proteoglycan with neurite-inhibiting activity is up-regulated following peripheral nerve injury. Journal of Neurobiology 34 (1), 41–54.

Zuo, J., Neubauer, D., Graham, J., Krekoski, C.A., Ferguson, T.A., Muir, D., 2002. Regeneration of axons after nerve transection repair is enhanced by degradation of chondroitin sulfate proteoglycan. Experimental Neurology 176 (1), 221–228.